Distributed Machine Learning Systems

Computer Assignment 3

Spark, HDFS, and Machine Learning

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Course report and execution evidence

Q1

I explained the codes in the .ipynb file.

Q2

I explained the codes in the .ipynb file.

Use this command to download the wiki_corpus:

```
scp -r karimi@dml0:/home/shared_files/CA2/wiki_corpus ~/Downloads/
```

Q3

1

This is how I put the “Titanic-Dataset.csv” dataset to the raspberrypi
server:

```
(base) alireza@alirezas-MacBook-Air ~ % scp ~/Documents/uni/term7/DMLS/CA/3/DMLS_CA3/Titanic-Dataset.csv
  karimi@dml0:~/CA3
```

I got this error while entering this command:

```
karimi@raspberrypi-dml0:~/CA3 $ hdfs dfs -mkdir -p /810101492
mkdir: Permission denied: user=karimi, access=WRITE, inode="/":hadoop:supergroup:drwxr-xr-x
```

And this is how i put the dataset to my folder:

```
karimi@raspberrypi-dml0:~ $ hdfs dfs -ls /
Found 22 items
drwxr-xr-x   - adib            dmls          0 2025-12-30 17:15 /adib
drwxr-xr-x   - afsari          dmls          0 2025-12-20 13:30 /afsari
drwxr-xr-x   - ahmadizarei     dmls          0 2026-01-04 05:44 /ahmadizarei
drwxr-xr-x   - amouei          dmls          0 2025-12-20 13:30 /amouei
drwxr-xr-x   - assarian        dmls          0 2025-12-20 13:30 /assarian
drwxr-xr-x   - bagherinia      dmls          0 2025-12-20 13:30 /bagherinia
drwxr-xr-x   - dousti          dmls          0 2025-12-20 14:25 /dousti
drwxr-xr-x   - ghertasi        dmls          0 2026-01-03 18:39 /ghertasi
drwxr-xr-x   - gholami         dmls          0 2025-12-29 17:05 /gholami
drwxr-xr-x   - hosseinnejad    dmls          0 2025-12-20 13:30 /hosseinnejad
drwxr-xr-x   - jafari          dmls          0 2025-12-29 11:35 /jafari
drwxr-xr-x   - jannati         dmls          0 2025-12-29 17:44 /jannati
drwxr-xr-x   - karimi          dmls          0 2025-12-20 13:30 /karimi
drwxr-xr-x   - khezri          dmls          0 2026-01-04 09:59 /khezri
drwxr-xr-x   - khoramfar       dmls          0 2025-12-20 13:30 /khoramfar
drwxr-xr-x   - massoudi        dmls          0 2025-12-20 13:30 /massoudi
drwxr-xr-x   - mehrabian       dmls          0 2026-01-01 10:41 /mehrabian
drwxr-xr-x   - mousavi         dmls          0 2025-12-20 13:30 /mousavi
drwxr-xr-x   - rahimi          dmls          0 2025-12-20 13:30 /rahimi
drwxr-xr-x   - ramezani        dmls          0 2025-12-20 15:39 /ramezani
drwxr-xr-x   - soltanmohammadi dmls          0 2026-01-02 23:47 /soltanmohammadi
drwxr-xr-x   - vajhi           dmls          0 2025-12-26 14:24 /vajhi
karimi@raspberrypi-dml0:~ $ hdfs dfs -ls /karimi
karimi@raspberrypi-dml0:~ $ hdfs dfs -put Titanic-Dataset.csv /karimi
put: `Titanic-Dataset.csv': No such file or directory
karimi@raspberrypi-dml0:~ $ hdfs dfs -ls /karimi
karimi@raspberrypi-dml0:~ $ hdfs dfs -put ~/CA3/Titanic-Dataset.csv /karimi
karimi@raspberrypi-dml0:~ $ hdfs dfs -ls /karimi
```

```
Found 1 items
-rw-r--r--   1 karimi dmls      61194 2026-01-04 14:01 /karimi/Titanic-Dataset.csv
karimi@raspberrypi-dml0:~ $
```

Got help from this:
https://gemini.google.com/share/609a508fdf57

2

```
karimi@raspberrypi-dml0:~/CA3 $ nano titanic_hdfs.py
karimi@raspberrypi-dml0:~/CA3 $ cat titanic_hdfs.py
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, avg, count, when

# 1. Initialize Spark Session
spark = SparkSession.builder \
    .appName("Titanic_Analysis_Karimi") \
    .getOrCreate()

# 2. Load Data from HDFS
# Based on your terminal output, the file is at /karimi/Titanic-Dataset.csv
hdfs_path = "hdfs://raspberrypi-dm10:9000/karimi/Titanic-Dataset.csv"

print(f"Loading data from: {hdfs_path} ...")
df = spark.read.csv(hdfs_path, header=True, inferSchema=True)

# 3. Calculate Survival Rate by Sex (Question 3.2 Part 1)
print("\n--- Survival Rate by Sex ---")
# Group by Sex, calculate average of 'Survived' (0 or 1)
# Since Survived is 0 or 1, the average * 100 gives the percentage.
sex_survival = df.groupBy("Sex").agg(
    (avg("Survived") * 100).alias("Survival_Rate_Percent"),
    count("PassengerId").alias("Total_Passengers")
)
sex_survival.show()

# 4. Calculate Survival Rate by Pclass (Question 3.2 Part 2)
print("\n--- Survival Rate by Ticket Class (Pclass) ---")
class_survival = df.groupBy("Pclass").agg(
    (avg("Survived") * 100).alias("Survival_Rate_Percent"),
    count("PassengerId").alias("Total_Passengers")
).orderBy("Pclass")
class_survival.show()

# 5. Average Age for Survivors vs Non-Survivors (Question 3.2 Part 3)
print("\n--- Average Age: Survivors vs Non-Survivors ---")
# 0 = No (Deceased), 1 = Yes (Survived)
age_analysis = df.groupBy("Survived").agg(
    avg("Age").alias("Average_Age"),
    count("Age").alias("Count_With_Age_Data") # Just to see how many had valid ages
)
age_analysis.show()

karimi@raspberrypi-dml0:~/CA3 $ spark-submit titanic_hdfs.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/01/04 14:25:21 INFO SparkContext: Running Spark version 4.1.0
26/01/04 14:25:21 INFO SparkContext: OS info Linux, 6.6.42-v8+, aarch64
26/01/04 14:25:21 INFO SparkContext: Java version 21.0.9+7-LTS-338
26/01/04 14:25:22 INFO ResourceUtils: ==============================================================
26/01/04 14:25:22 INFO ResourceUtils: No custom resources configured for spark.driver.
26/01/04 14:25:22 INFO ResourceUtils: ==============================================================
26/01/04 14:25:22 INFO SparkContext: Submitted application: Titanic_Analysis_Karimi
26/01/04 14:25:22 INFO SecurityManager: Changing view acls to: hadoop,dousti,karimi
26/01/04 14:25:22 INFO SecurityManager: Changing modify acls to: hadoop,dousti,karimi
26/01/04 14:25:22 INFO SecurityManager: Changing view acls groups to: hadoop,dousti,karimi
26/01/04 14:25:22 INFO SecurityManager: Changing modify acls groups to: hadoop,dousti,karimi
26/01/04 14:25:22 INFO SecurityManager: SecurityManager: authentication disabled; ui acls enabled; users with
  view permissions: hadoop, dousti, karimi groups with view permissions: EMPTY; users with modify permissions:
  hadoop, dousti, karimi; groups with modify permissions: EMPTY; RPC SSL disabled
26/01/04 14:25:24 INFO Utils: Successfully started service 'sparkDriver' on port 42469.
26/01/04 14:25:24 INFO SparkEnv: Registering MapOutputTracker
26/01/04 14:25:24 INFO SparkEnv: Registering BlockManagerMaster
26/01/04 14:25:24 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for
  getting topology information
26/01/04 14:25:24 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/01/04 14:25:25 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
```

```
26/01/04 14:25:25 INFO DiskBlockManager: Created local directory at
  /tmp/blockmgr-bd61f5b0-1143-450c-ac13-63b8d0757324
26/01/04 14:25:25 INFO SparkEnv: Registering OutputCommitCoordinator
26/01/04 14:25:26 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI
26/01/04 14:25:26 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/01/04 14:25:26 INFO ServerInfo: Adding filter to /: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:27 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name:
   cores, amount: 2, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap ->
  name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/01/04 14:25:27 INFO ResourceProfile: Limiting resource is cpus at 2 tasks per executor
26/01/04 14:25:27 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/01/04 14:25:27 INFO SecurityManager: Changing view acls to: hadoop,dousti,karimi
26/01/04 14:25:27 INFO SecurityManager: Changing modify acls to: hadoop,dousti,karimi
26/01/04 14:25:27 INFO SecurityManager: Changing view acls groups to: hadoop,dousti,karimi
26/01/04 14:25:27 INFO SecurityManager: Changing modify acls groups to: hadoop,dousti,karimi
26/01/04 14:25:27 INFO SecurityManager: SecurityManager: authentication disabled; ui acls enabled; users with
  view permissions: hadoop, dousti, karimi groups with view permissions: EMPTY; users with modify permissions:
  hadoop, dousti, karimi; groups with modify permissions: EMPTY; RPC SSL disabled
26/01/04 14:25:28 INFO StandaloneAppClient$ClientEndpoint: Connecting to master spark://172.18.32.200:7077...
26/01/04 14:25:28 INFO TransportClientFactory: Successfully created connection to /172.18.32.200:7077 after 186
   ms (0 ms spent in bootstraps)
26/01/04 14:25:29 INFO StandaloneSchedulerBackend: Connected to Spark cluster with app ID
  app-20260104142529-0030
26/01/04 14:25:29 INFO StandaloneAppClient$ClientEndpoint: Executor added: app-20260104142529-0030/0 on
  worker-20260104001835-172.18.32.203-41527 (172.18.32.203:41527) with 2 core(s)
26/01/04 14:25:29 INFO StandaloneSchedulerBackend: Granted executor ID app-20260104142529-0030/0 on hostPort
  172.18.32.203:41527 with 2 core(s), 1024.0 MiB RAM
26/01/04 14:25:29 INFO StandaloneAppClient$ClientEndpoint: Executor added: app-20260104142529-0030/1 on
  worker-20260104001832-172.18.32.200-35083 (172.18.32.200:35083) with 2 core(s)
26/01/04 14:25:29 INFO StandaloneSchedulerBackend: Granted executor ID app-20260104142529-0030/1 on hostPort
  172.18.32.200:35083 with 2 core(s), 1024.0 MiB RAM
26/01/04 14:25:29 INFO Utils: Successfully started service
  'org.apache.spark.network.netty.NettyBlockTransferService' on port 43563.
26/01/04 14:25:29 INFO NettyBlockTransferService: Server created on raspberrypi-dml0:43563
26/01/04 14:25:29 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block
  replication policy
26/01/04 14:25:29 INFO StandaloneAppClient$ClientEndpoint: Executor updated: app-20260104142529-0030/1 is now
  RUNNING
26/01/04 14:25:29 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, raspberrypi-dml0,
  43563, None)
26/01/04 14:25:29 INFO BlockManagerMasterEndpoint: Registering block manager raspberrypi-dml0:43563 with 434.4
  MiB RAM, BlockManagerId(driver, raspberrypi-dml0, 43563, None)
26/01/04 14:25:30 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, raspberrypi-dml0,
  43563, None)
26/01/04 14:25:30 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, raspberrypi-dml0, 43563,
  None)
26/01/04 14:25:30 INFO StandaloneAppClient$ClientEndpoint: Executor updated: app-20260104142529-0030/0 is now
  RUNNING
26/01/04 14:25:30 INFO ServerInfo: Adding filter to /jobs: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:30 INFO ServerInfo: Adding filter to /jobs/json: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:30 INFO ServerInfo: Adding filter to /jobs/job: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:30 INFO ServerInfo: Adding filter to /jobs/job/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:30 INFO ServerInfo: Adding filter to /stages: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:30 INFO ServerInfo: Adding filter to /stages/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /stages/stage:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /stages/stage/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /stages/pool:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /stages/pool/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /stages/taskThreadDump:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /stages/taskThreadDump/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /storage: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /storage/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /storage/rdd:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /storage/rdd/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /environment:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /environment/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /executors: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /executors/json:
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  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /executors/threadDump:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /executors/threadDump/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /executors/heapHistogram:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /executors/heapHistogram/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /static: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /api: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /metrics: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /jobs/job/kill:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /stages/stage/kill:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO ServerInfo: Adding filter to /metrics/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:31 INFO StandaloneSchedulerBackend: SchedulerBackend is ready for scheduling beginning after
  reached minRegisteredResourcesRatio: 0.0
26/01/04 14:25:42 INFO StandaloneSchedulerBackend$StandaloneDriverEndpoint: Registered executor
  NettyRpcEndpointRef(spark-client://Executor) (172.18.32.200:35884) with ID 1, ResourceProfileId 0
26/01/04 14:25:43 INFO BlockManagerMasterEndpoint: Registering block manager 172.18.32.200:38381 with 434.4 MiB
   RAM, BlockManagerId(1, 172.18.32.200, 38381, None)
Loading data from: hdfs://raspberrypi-dm10:9000/karimi/Titanic-Dataset.csv ...
26/01/04 14:25:47 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of
  spark.sql.warehouse.dir.
26/01/04 14:25:48 INFO SharedState: Warehouse path is 'file:/home/dmls/karimi/CA3/spark-warehouse'.
26/01/04 14:25:48 INFO ServerInfo: Adding filter to /SQL: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:48 INFO ServerInfo: Adding filter to /SQL/json: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:48 INFO ServerInfo: Adding filter to /SQL/execution:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:48 INFO ServerInfo: Adding filter to /SQL/execution/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:48 INFO ServerInfo: Adding filter to /static/sql:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:25:48 INFO StandaloneSchedulerBackend$StandaloneDriverEndpoint: Registered executor
  NettyRpcEndpointRef(spark-client://Executor) (172.18.32.203:49612) with ID 0, ResourceProfileId 0
26/01/04 14:25:50 INFO BlockManagerMasterEndpoint: Registering block manager 172.18.32.203:44839 with 434.4 MiB
   RAM, BlockManagerId(0, 172.18.32.203, 44839, None)
26/01/04 14:25:51 WARN FileSystem: Failed to initialize filesystem
  hdfs://raspberrypi-dm10:9000/karimi/Titanic-Dataset.csv: java.lang.IllegalArgumentException:
  java.net.UnknownHostException: raspberrypi-dm10
26/01/04 14:25:51 WARN FileStreamSink: Assume no metadata directory. Error while looking for metadata directory
   in the path: hdfs://raspberrypi-dm10:9000/karimi/Titanic-Dataset.csv.
java.lang.IllegalArgumentException: java.net.UnknownHostException: raspberrypi-dm10
    at org.apache.hadoop.security.SecurityUtil.buildTokenService(SecurityUtil.java:479)
    at
  org.apache.hadoop.hdfs.NameNodeProxiesClient.createProxyWithClientProtocol(NameNodeProxiesClient.java:134)
    at org.apache.hadoop.hdfs.DFSClient.<init>(DFSClient.java:370)
    at org.apache.hadoop.hdfs.DFSClient.<init>(DFSClient.java:309)
    at org.apache.hadoop.hdfs.DistributedFileSystem.initDFSClient(DistributedFileSystem.java:205)
    at org.apache.hadoop.hdfs.DistributedFileSystem.initialize(DistributedFileSystem.java:190)
    at org.apache.hadoop.fs.FileSystem.createFileSystem(FileSystem.java:3615)
    at org.apache.hadoop.fs.FileSystem.access$300(FileSystem.java:172)
    at org.apache.hadoop.fs.FileSystem$Cache.getInternal(FileSystem.java:3716)
    at org.apache.hadoop.fs.FileSystem$Cache.get(FileSystem.java:3667)
    at org.apache.hadoop.fs.FileSystem.get(FileSystem.java:557)
    at org.apache.hadoop.fs.Path.getFileSystem(Path.java:366)
    at org.apache.spark.sql.execution.streaming.sinks.FileStreamSink$.hasMetadata(FileStreamSink.scala:57)
    at org.apache.spark.sql.execution.datasources.DataSource.resolveRelation(DataSource.scala:384)
    at org.apache.spark.sql.catalyst.analysis.ResolveDataSource.org$apache$spark$sql$catalyst$analysis$ResolveD
  ataSource$$loadV1BatchSource(ResolveDataSource.scala:143)
    at org.apache.spark.sql.catalyst.analysis.ResolveDataSource$$anonfun$apply$1.$anonfun$applyOrElse$2(Resolve
  DataSource.scala:61)
    at scala.Option.getOrElse(Option.scala:201)
    at org.apache.spark.sql.catalyst.analysis.ResolveDataSource$$anonfun$apply$1.applyOrElse(ResolveDataSource.
  scala:61)
    at org.apache.spark.sql.catalyst.analysis.ResolveDataSource$$anonfun$apply$1.applyOrElse(ResolveDataSource.
  scala:45)
    at org.apache.spark.sql.catalyst.plans.logical.AnalysisHelper.$anonfun$resolveOperatorsUpWithPruning$3(Anal
  ysisHelper.scala:139)
    at org.apache.spark.sql.catalyst.trees.CurrentOrigin$.withOrigin(origin.scala:107)
    at org.apache.spark.sql.catalyst.plans.logical.AnalysisHelper.$anonfun$resolveOperatorsUpWithPruning$1(Anal
  ysisHelper.scala:139)
    at org.apache.spark.sql.catalyst.plans.logical.AnalysisHelper$.allowInvokingTransformsInAnalyzer(AnalysisHe
  lper.scala:416)
    at org.apache.spark.sql.catalyst.plans.logical.AnalysisHelper.resolveOperatorsUpWithPruning(AnalysisHelper.
  scala:135)
    at org.apache.spark.sql.catalyst.plans.logical.AnalysisHelper.resolveOperatorsUpWithPruning$(AnalysisHelper
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  .scala:131)
    at
  org.apache.spark.sql.catalyst.plans.logical.LogicalPlan.resolveOperatorsUpWithPruning(LogicalPlan.scala:37)
    at org.apache.spark.sql.catalyst.plans.logical.AnalysisHelper.resolveOperatorsUp(AnalysisHelper.scala:112)
    at org.apache.spark.sql.catalyst.plans.logical.AnalysisHelper.resolveOperatorsUp$(AnalysisHelper.scala:111)
    at org.apache.spark.sql.catalyst.plans.logical.LogicalPlan.resolveOperatorsUp(LogicalPlan.scala:37)
    at org.apache.spark.sql.catalyst.analysis.ResolveDataSource.apply(ResolveDataSource.scala:45)
    at org.apache.spark.sql.catalyst.analysis.ResolveDataSource.apply(ResolveDataSource.scala:43)
    at org.apache.spark.sql.catalyst.rules.RuleExecutor.$anonfun$execute$2(RuleExecutor.scala:248)
    at scala.collection.LinearSeqOps.foldLeft(LinearSeq.scala:183)
    at scala.collection.LinearSeqOps.foldLeft$(LinearSeq.scala:179)
    at scala.collection.immutable.List.foldLeft(List.scala:79)
    at org.apache.spark.sql.catalyst.rules.RuleExecutor.$anonfun$execute$1(RuleExecutor.scala:245)
    at org.apache.spark.sql.catalyst.rules.RuleExecutor.$anonfun$execute$1$adapted(RuleExecutor.scala:237)
    at scala.collection.immutable.List.foreach(List.scala:323)
    at org.apache.spark.sql.catalyst.rules.RuleExecutor.execute(RuleExecutor.scala:237)
    at org.apache.spark.sql.catalyst.analysis.Analyzer.org$apache$spark$sql$catalyst$analysis$Analyzer$$execute
  SameContext(Analyzer.scala:343)
    at org.apache.spark.sql.catalyst.analysis.Analyzer.$anonfun$execute$1(Analyzer.scala:339)
    at org.apache.spark.sql.catalyst.analysis.AnalysisContext$.withNewAnalysisContext(Analyzer.scala:224)
    at org.apache.spark.sql.catalyst.analysis.Analyzer.execute(Analyzer.scala:339)
    at org.apache.spark.sql.catalyst.analysis.Analyzer.execute(Analyzer.scala:289)
    at org.apache.spark.sql.catalyst.rules.RuleExecutor.$anonfun$executeAndTrack$1(RuleExecutor.scala:207)
    at org.apache.spark.sql.catalyst.QueryPlanningTracker$.withTracker(QueryPlanningTracker.scala:89)
    at org.apache.spark.sql.catalyst.rules.RuleExecutor.executeAndTrack(RuleExecutor.scala:207)
    at
  org.apache.spark.sql.catalyst.analysis.resolver.HybridAnalyzer.resolveInFixedPoint(HybridAnalyzer.scala:236)
    at org.apache.spark.sql.catalyst.analysis.resolver.HybridAnalyzer.$anonfun$apply$1(HybridAnalyzer.scala:91)
    at org.apache.spark.sql.catalyst.analysis.resolver.HybridAnalyzer.withTrackedAnalyzerBridgeState(HybridAnal
  yzer.scala:122)
    at org.apache.spark.sql.catalyst.analysis.resolver.HybridAnalyzer.apply(HybridAnalyzer.scala:84)
    at org.apache.spark.sql.catalyst.analysis.Analyzer.$anonfun$executeAndCheck$1(Analyzer.scala:322)
    at org.apache.spark.sql.catalyst.plans.logical.AnalysisHelper$.markInAnalyzer(AnalysisHelper.scala:423)
    at org.apache.spark.sql.catalyst.analysis.Analyzer.executeAndCheck(Analyzer.scala:322)
    at org.apache.spark.sql.execution.QueryExecution.$anonfun$lazyAnalyzed$2(QueryExecution.scala:139)
    at org.apache.spark.sql.catalyst.QueryPlanningTracker.measurePhase(QueryPlanningTracker.scala:148)
    at org.apache.spark.sql.execution.QueryExecution.$anonfun$executePhase$2(QueryExecution.scala:330)
    at org.apache.spark.sql.execution.QueryExecution$.withInternalError(QueryExecution.scala:717)
    at org.apache.spark.sql.execution.QueryExecution.$anonfun$executePhase$1(QueryExecution.scala:330)
    at org.apache.spark.sql.SparkSession.withActive(SparkSession.scala:804)
    at org.apache.spark.sql.execution.QueryExecution.executePhase(QueryExecution.scala:329)
    at org.apache.spark.sql.execution.QueryExecution.$anonfun$lazyAnalyzed$1(QueryExecution.scala:139)
    at scala.util.Try$.apply(Try.scala:217)
    at org.apache.spark.util.Utils$.doTryWithCallerStacktrace(Utils.scala:1392)
    at org.apache.spark.util.LazyTry.tryT$lzycompute(LazyTry.scala:46)
    at org.apache.spark.util.LazyTry.tryT(LazyTry.scala:46)
    at org.apache.spark.util.LazyTry.get(LazyTry.scala:58)
    at org.apache.spark.sql.execution.QueryExecution.analyzed(QueryExecution.scala:150)
    at org.apache.spark.sql.execution.QueryExecution.assertAnalyzed(QueryExecution.scala:90)
    at org.apache.spark.sql.classic.Dataset$.$anonfun$ofRows$1(Dataset.scala:114)
    at org.apache.spark.sql.SparkSession.withActive(SparkSession.scala:804)
    at org.apache.spark.sql.classic.Dataset$.ofRows(Dataset.scala:112)
    at org.apache.spark.sql.classic.DataFrameReader.load(DataFrameReader.scala:108)
    at org.apache.spark.sql.classic.DataFrameReader.load(DataFrameReader.scala:57)
    at org.apache.spark.sql.DataFrameReader.csv(DataFrameReader.scala:392)
    at org.apache.spark.sql.classic.DataFrameReader.csv(DataFrameReader.scala:258)
    at org.apache.spark.sql.classic.DataFrameReader.csv(DataFrameReader.scala:57)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:75)
    at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:52)
    at java.base/java.lang.reflect.Method.invoke(Method.java:580)
    at py4j.reflection.MethodInvoker.invoke(MethodInvoker.java:244)
    at py4j.reflection.ReflectionEngine.invoke(ReflectionEngine.java:374)
    at py4j.Gateway.invoke(Gateway.java:282)
    at py4j.commands.AbstractCommand.invokeMethod(AbstractCommand.java:132)
    at py4j.commands.CallCommand.execute(CallCommand.java:79)
    at py4j.ClientServerConnection.waitForCommands(ClientServerConnection.java:184)
    at py4j.ClientServerConnection.run(ClientServerConnection.java:108)
    at java.base/java.lang.Thread.run(Thread.java:1583)
Caused by: java.net.UnknownHostException: raspberrypi-dm10
    ... 89 more
26/01/04 14:25:51 WARN FileSystem: Failed to initialize filesystem
  hdfs://raspberrypi-dm10:9000/karimi/Titanic-Dataset.csv: java.lang.IllegalArgumentException:
  java.net.UnknownHostException: raspberrypi-dm10
Traceback (most recent call last):
  File "/home/dmls/karimi/CA3/titanic_hdfs.py", line 14, in <module>
    df = spark.read.csv(hdfs_path, header=True, inferSchema=True)
  File "/home/shared_files/spark/python/lib/pyspark.zip/pyspark/sql/readwriter.py", line 838, in csv
    return self._df(self._jreader.csv(self._spark._sc._jvm.PythonUtils.toSeq(path)))
                    ~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/shared_files/spark/python/lib/py4j-0.10.9.9-src.zip/py4j/java_gateway.py", line 1362, in __call__
```

```
    return_value = get_return_value(
        answer, self.gateway_client, self.target_id, self.name)
  File "/home/shared_files/spark/python/lib/pyspark.zip/pyspark/errors/exceptions/captured.py", line 269, in
  deco
    raise converted from None
pyspark.errors.exceptions.captured.IllegalArgumentException: java.net.UnknownHostException: raspberrypi-dm10
26/01/04 14:25:52 INFO SparkContext: Invoking stop() from shutdown hook
26/01/04 14:25:52 INFO SparkContext: SparkContext is stopping with exitCode 0 from run at Executors.java:572.
26/01/04 14:25:52 INFO SparkUI: Stopped Spark web UI at http://raspberrypi-dml0:4040
26/01/04 14:25:52 INFO StandaloneSchedulerBackend: Shutting down all executors
26/01/04 14:25:52 INFO StandaloneSchedulerBackend$StandaloneDriverEndpoint: Asking each executor to shut down
26/01/04 14:25:52 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/01/04 14:25:52 INFO MemoryStore: MemoryStore started with capacity 434.4 MiB
26/01/04 14:25:52 INFO MemoryStore: MemoryStore cleared
26/01/04 14:25:52 INFO BlockManager: BlockManager stopped
26/01/04 14:25:52 INFO BlockManagerMaster: BlockManagerMaster stopped
26/01/04 14:25:52 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator
  stopped!
26/01/04 14:25:52 INFO SparkContext: Successfully stopped SparkContext (Uptime: 32079 ms)
karimi@raspberrypi-dml0:~/CA3 $
```

I have mistakenly used “raspberrypi-dm10” instead of “dml0”.

So I editted the code:

```
karimi@raspberrypi-dml0:~/CA3 $ nano titanic_hdfs.py
karimi@raspberrypi-dml0:~/CA3 $ cat titanic_hdfs.py
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, avg, count, when

# 1. Initialize Spark Session
spark = SparkSession.builder \
    .appName("Titanic_Analysis_Karimi") \
    .getOrCreate()

# 2. Load Data from HDFS
# Based on your terminal output, the file is at /karimi/Titanic-Dataset.csv
hdfs_path = "hdfs://172.18.32.200:9000/karimi/Titanic-Dataset.csv"

print(f"Loading data from: {hdfs_path} ...")
df = spark.read.csv(hdfs_path, header=True, inferSchema=True)

# 3. Calculate Survival Rate by Sex (Question 3.2 Part 1)
print("\n--- Survival Rate by Sex ---")
# Group by Sex, calculate average of 'Survived' (0 or 1)
# Since Survived is 0 or 1, the average * 100 gives the percentage.
sex_survival = df.groupBy("Sex").agg(
    (avg("Survived") * 100).alias("Survival_Rate_Percent"),
    count("PassengerId").alias("Total_Passengers")
)
sex_survival.show()

# 4. Calculate Survival Rate by Pclass (Question 3.2 Part 2)
print("\n--- Survival Rate by Ticket Class (Pclass) ---")
class_survival = df.groupBy("Pclass").agg(
    (avg("Survived") * 100).alias("Survival_Rate_Percent"),
    count("PassengerId").alias("Total_Passengers")
).orderBy("Pclass")
class_survival.show()

# 5. Average Age for Survivors vs Non-Survivors (Question 3.2 Part 3)
print("\n--- Average Age: Survivors vs Non-Survivors ---")
# 0 = No (Deceased), 1 = Yes (Survived)
age_analysis = df.groupBy("Survived").agg(
    avg("Age").alias("Average_Age"),
    count("Age").alias("Count_With_Age_Data") # Just to see how many had valid ages
)
age_analysis.show()

karimi@raspberrypi-dml0:~/CA3 $ spark-submit titanic_hdfs.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/01/04 14:34:14 INFO SparkContext: Running Spark version 4.1.0
26/01/04 14:34:14 INFO SparkContext: OS info Linux, 6.6.42-v8+, aarch64
26/01/04 14:34:14 INFO SparkContext: Java version 21.0.9+7-LTS-338
26/01/04 14:34:15 INFO ResourceUtils: ==============================================================
26/01/04 14:34:15 INFO ResourceUtils: No custom resources configured for spark.driver.
26/01/04 14:34:15 INFO ResourceUtils: ==============================================================
26/01/04 14:34:15 INFO SparkContext: Submitted application: Titanic_Analysis_Karimi
```

```
26/01/04 14:34:15 INFO SecurityManager: Changing view acls to: hadoop,dousti,karimi
26/01/04 14:34:15 INFO SecurityManager: Changing modify acls to: hadoop,dousti,karimi
26/01/04 14:34:15 INFO SecurityManager: Changing view acls groups to: hadoop,dousti,karimi
26/01/04 14:34:15 INFO SecurityManager: Changing modify acls groups to: hadoop,dousti,karimi
26/01/04 14:34:15 INFO SecurityManager: SecurityManager: authentication disabled; ui acls enabled; users with
  view permissions: hadoop, dousti, karimi groups with view permissions: EMPTY; users with modify permissions:
  hadoop, dousti, karimi; groups with modify permissions: EMPTY; RPC SSL disabled
26/01/04 14:34:16 INFO Utils: Successfully started service 'sparkDriver' on port 33721.
26/01/04 14:34:16 INFO SparkEnv: Registering MapOutputTracker
26/01/04 14:34:16 INFO SparkEnv: Registering BlockManagerMaster
26/01/04 14:34:16 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for
  getting topology information
26/01/04 14:34:16 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/01/04 14:34:16 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/01/04 14:34:16 INFO DiskBlockManager: Created local directory at
  /tmp/blockmgr-467e304a-1da7-4dc9-b5d0-1375211995df
26/01/04 14:34:17 INFO SparkEnv: Registering OutputCommitCoordinator
26/01/04 14:34:17 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI
26/01/04 14:34:17 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/01/04 14:34:17 INFO ServerInfo: Adding filter to /: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:18 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name:
   cores, amount: 2, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap ->
  name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/01/04 14:34:18 INFO ResourceProfile: Limiting resource is cpus at 2 tasks per executor
26/01/04 14:34:18 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/01/04 14:34:18 INFO SecurityManager: Changing view acls to: hadoop,dousti,karimi
26/01/04 14:34:18 INFO SecurityManager: Changing modify acls to: hadoop,dousti,karimi
26/01/04 14:34:18 INFO SecurityManager: Changing view acls groups to: hadoop,dousti,karimi
26/01/04 14:34:18 INFO SecurityManager: Changing modify acls groups to: hadoop,dousti,karimi
26/01/04 14:34:18 INFO SecurityManager: SecurityManager: authentication disabled; ui acls enabled; users with
  view permissions: hadoop, dousti, karimi groups with view permissions: EMPTY; users with modify permissions:
  hadoop, dousti, karimi; groups with modify permissions: EMPTY; RPC SSL disabled
26/01/04 14:34:18 INFO StandaloneAppClient$ClientEndpoint: Connecting to master spark://172.18.32.200:7077...
26/01/04 14:34:19 INFO TransportClientFactory: Successfully created connection to /172.18.32.200:7077 after 151
   ms (0 ms spent in bootstraps)
26/01/04 14:34:19 INFO StandaloneSchedulerBackend: Connected to Spark cluster with app ID
  app-20260104143419-0032
26/01/04 14:34:19 INFO StandaloneAppClient$ClientEndpoint: Executor added: app-20260104143419-0032/0 on
  worker-20260104001835-172.18.32.203-41527 (172.18.32.203:41527) with 2 core(s)
26/01/04 14:34:19 INFO StandaloneSchedulerBackend: Granted executor ID app-20260104143419-0032/0 on hostPort
  172.18.32.203:41527 with 2 core(s), 1024.0 MiB RAM
26/01/04 14:34:19 INFO StandaloneAppClient$ClientEndpoint: Executor added: app-20260104143419-0032/1 on
  worker-20260104001832-172.18.32.200-35083 (172.18.32.200:35083) with 2 core(s)
26/01/04 14:34:19 INFO StandaloneSchedulerBackend: Granted executor ID app-20260104143419-0032/1 on hostPort
  172.18.32.200:35083 with 2 core(s), 1024.0 MiB RAM
26/01/04 14:34:19 INFO Utils: Successfully started service
  'org.apache.spark.network.netty.NettyBlockTransferService' on port 38861.
26/01/04 14:34:19 INFO NettyBlockTransferService: Server created on raspberrypi-dml0:38861
26/01/04 14:34:19 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block
  replication policy
26/01/04 14:34:20 INFO StandaloneAppClient$ClientEndpoint: Executor updated: app-20260104143419-0032/1 is now
  RUNNING
26/01/04 14:34:20 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, raspberrypi-dml0,
  38861, None)
26/01/04 14:34:20 INFO BlockManagerMasterEndpoint: Registering block manager raspberrypi-dml0:38861 with 434.4
  MiB RAM, BlockManagerId(driver, raspberrypi-dml0, 38861, None)
26/01/04 14:34:20 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, raspberrypi-dml0,
  38861, None)
26/01/04 14:34:20 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, raspberrypi-dml0, 38861,
  None)
26/01/04 14:34:20 INFO StandaloneAppClient$ClientEndpoint: Executor updated: app-20260104143419-0032/0 is now
  RUNNING
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /jobs: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /jobs/json: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /jobs/job: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /jobs/job/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /stages: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /stages/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /stages/stage:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /stages/stage/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /stages/pool:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /stages/pool/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /stages/taskThreadDump:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /stages/taskThreadDump/json:
```

```
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /storage: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /storage/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /storage/rdd:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /storage/rdd/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /environment:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /environment/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:20 INFO ServerInfo: Adding filter to /executors: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /executors/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /executors/threadDump:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /executors/threadDump/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /executors/heapHistogram:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /executors/heapHistogram/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /static: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /api: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /metrics: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /jobs/job/kill:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /stages/stage/kill:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO ServerInfo: Adding filter to /metrics/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:21 INFO StandaloneSchedulerBackend: SchedulerBackend is ready for scheduling beginning after
  reached minRegisteredResourcesRatio: 0.0
26/01/04 14:34:34 INFO StandaloneSchedulerBackend$StandaloneDriverEndpoint: Registered executor
  NettyRpcEndpointRef(spark-client://Executor) (172.18.32.200:56666) with ID 1, ResourceProfileId 0
26/01/04 14:34:35 INFO BlockManagerMasterEndpoint: Registering block manager 172.18.32.200:42119 with 434.4 MiB
   RAM, BlockManagerId(1, 172.18.32.200, 42119, None)
Loading data from: hdfs://172.18.32.200:9000/karimi/Titanic-Dataset.csv ...
26/01/04 14:34:36 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of
  spark.sql.warehouse.dir.
26/01/04 14:34:36 INFO SharedState: Warehouse path is 'file:/home/dmls/karimi/CA3/spark-warehouse'.
26/01/04 14:34:36 INFO ServerInfo: Adding filter to /SQL: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:36 INFO ServerInfo: Adding filter to /SQL/json: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:36 INFO ServerInfo: Adding filter to /SQL/execution:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:36 INFO ServerInfo: Adding filter to /SQL/execution/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:36 INFO ServerInfo: Adding filter to /static/sql:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:34:39 INFO StandaloneSchedulerBackend$StandaloneDriverEndpoint: Registered executor
  NettyRpcEndpointRef(spark-client://Executor) (172.18.32.203:50888) with ID 0, ResourceProfileId 0
26/01/04 14:34:40 INFO BlockManagerMasterEndpoint: Registering block manager 172.18.32.203:45167 with 434.4 MiB
   RAM, BlockManagerId(0, 172.18.32.203, 45167, None)
26/01/04 14:34:41 INFO InMemoryFileIndex: It took 298 ms to list leaf files for 1 paths.
26/01/04 14:34:42 INFO InMemoryFileIndex: It took 16 ms to list leaf files for 1 paths.
26/01/04 14:34:45 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:34:46 INFO FileSourceStrategy: Post-Scan Filters: Set((length(trim(value#0, None)) > 0))
26/01/04 14:34:48 INFO CodeGenerator: Code generated in 956.497871 ms
26/01/04 14:34:48 INFO MemoryStore: MemoryStore started with capacity 434.4 MiB
26/01/04 14:34:48 INFO MemoryStore: Block broadcast_0 stored as values in memory (estimated size 215.6 KiB,
  free 434.2 MiB)
26/01/04 14:34:48 INFO MemoryStore: Block broadcast_0_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 434.2 MiB)
26/01/04 14:34:48 INFO SparkContext: Created broadcast 0 from csv at NativeMethodAccessorImpl.java:0
26/01/04 14:34:48 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:34:49 INFO SparkContext: Starting job: csv at NativeMethodAccessorImpl.java:0
26/01/04 14:34:49 INFO DAGScheduler: Got job 0 (csv at NativeMethodAccessorImpl.java:0) with 1 output
  partitions
26/01/04 14:34:49 INFO DAGScheduler: Final stage: ResultStage 0 (csv at NativeMethodAccessorImpl.java:0)
26/01/04 14:34:49 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:34:49 INFO DAGScheduler: Missing parents: List()
26/01/04 14:34:49 INFO DAGScheduler: Missing parents found for ResultStage 0: List()
26/01/04 14:34:49 INFO DAGScheduler: Submitting ResultStage 0 (MapPartitionsRDD[3] at csv at
  NativeMethodAccessorImpl.java:0), which has no missing parents
26/01/04 14:34:49 INFO MemoryStore: Block broadcast_1 stored as values in memory (estimated size 14.3 KiB, free
   434.1 MiB)
26/01/04 14:34:49 INFO MemoryStore: Block broadcast_1_piece0 stored as bytes in memory (estimated size 6.8 KiB,
   free 434.1 MiB)
```

```
26/01/04 14:34:49 INFO SparkContext: Created broadcast 1 from broadcast at DAGScheduler.scala:1701
26/01/04 14:34:49 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 0 (MapPartitionsRDD[3] at csv
  at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
26/01/04 14:34:49 INFO TaskSchedulerImpl: Adding task set 0.0 with 1 tasks resource profile 0
26/01/04 14:34:49 INFO TaskSetManager: Starting task 0.0 in stage 0.0 (TID 0) (172.18.32.200,executor 1,
  partition 0, ANY, 10250 bytes)
26/01/04 14:34:57 INFO TaskSetManager: Finished task 0.0 in stage 0.0 (TID 0) in 7271 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:34:57 INFO TaskSchedulerImpl: Removed TaskSet 0.0 whose tasks have all completed, from pool
26/01/04 14:34:57 INFO DAGScheduler: ResultStage 0 (csv at NativeMethodAccessorImpl.java:0) finished in 7687 ms
26/01/04 14:34:57 INFO DAGScheduler: Job 0 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:34:57 INFO TaskSchedulerImpl: Canceling stage 0
26/01/04 14:34:57 INFO TaskSchedulerImpl: Killing all running tasks in stage 0: Stage finished
26/01/04 14:34:57 INFO DAGScheduler: Job 0 finished: csv at NativeMethodAccessorImpl.java:0, took 7842.66604 ms
26/01/04 14:34:57 INFO CodeGenerator: Code generated in 58.331826 ms
26/01/04 14:34:57 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:34:57 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/01/04 14:34:57 INFO MemoryStore: Block broadcast_2 stored as values in memory (estimated size 215.6 KiB,
  free 433.9 MiB)
26/01/04 14:34:57 INFO MemoryStore: Block broadcast_2_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.9 MiB)
26/01/04 14:34:57 INFO SparkContext: Created broadcast 2 from csv at NativeMethodAccessorImpl.java:0
26/01/04 14:34:57 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:34:58 INFO SparkContext: Starting job: csv at NativeMethodAccessorImpl.java:0
26/01/04 14:34:58 INFO DAGScheduler: Got job 1 (csv at NativeMethodAccessorImpl.java:0) with 1 output
  partitions
26/01/04 14:34:58 INFO DAGScheduler: Final stage: ResultStage 1 (csv at NativeMethodAccessorImpl.java:0)
26/01/04 14:34:58 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:34:58 INFO DAGScheduler: Missing parents: List()
26/01/04 14:34:58 INFO DAGScheduler: Missing parents found for ResultStage 1: List()
26/01/04 14:34:58 INFO DAGScheduler: Submitting ResultStage 1 (MapPartitionsRDD[9] at csv at
  NativeMethodAccessorImpl.java:0), which has no missing parents
26/01/04 14:34:58 INFO MemoryStore: Block broadcast_3 stored as values in memory (estimated size 23.4 KiB, free
   434.1 MiB)
26/01/04 14:34:58 INFO MemoryStore: Block broadcast_3_piece0 stored as bytes in memory (estimated size 10.7
  KiB, free 434.1 MiB)
26/01/04 14:34:58 INFO SparkContext: Created broadcast 3 from broadcast at DAGScheduler.scala:1701
26/01/04 14:34:58 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 1 (MapPartitionsRDD[9] at csv
  at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
26/01/04 14:34:58 INFO TaskSchedulerImpl: Adding task set 1.0 with 1 tasks resource profile 0
26/01/04 14:34:58 INFO TaskSetManager: Starting task 0.0 in stage 1.0 (TID 1) (172.18.32.200,executor 1,
  partition 0, ANY, 10250 bytes)
26/01/04 14:35:04 INFO TaskSetManager: Finished task 0.0 in stage 1.0 (TID 1) in 5599 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:35:04 INFO TaskSchedulerImpl: Removed TaskSet 1.0 whose tasks have all completed, from pool
26/01/04 14:35:04 INFO DAGScheduler: ResultStage 1 (csv at NativeMethodAccessorImpl.java:0) finished in 5770 ms
26/01/04 14:35:04 INFO DAGScheduler: Job 1 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:35:04 INFO TaskSchedulerImpl: Canceling stage 1
26/01/04 14:35:04 INFO TaskSchedulerImpl: Killing all running tasks in stage 1: Stage finished
26/01/04 14:35:04 INFO DAGScheduler: Job 1 finished: csv at NativeMethodAccessorImpl.java:0, took 5818.841714
  ms

--- Survival Rate by Sex ---
26/01/04 14:35:09 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:35:09 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/01/04 14:35:10 INFO CodeGenerator: Code generated in 336.505452 ms
26/01/04 14:35:10 INFO MemoryStore: Block broadcast_4 stored as values in memory (estimated size 215.5 KiB,
  free 433.9 MiB)
26/01/04 14:35:10 INFO MemoryStore: Block broadcast_4_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.9 MiB)
26/01/04 14:35:10 INFO SparkContext: Created broadcast 4 from $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:35:10 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:35:10 INFO DAGScheduler: Registering RDD 13 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 0
26/01/04 14:35:10 INFO DAGScheduler: Got map stage job 2 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:35:10 INFO DAGScheduler: Final stage: ShuffleMapStage 2 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:35:10 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:35:10 INFO DAGScheduler: Missing parents: List()
26/01/04 14:35:10 INFO DAGScheduler: Missing parents found for ShuffleMapStage 2: List()
26/01/04 14:35:10 INFO DAGScheduler: Submitting ShuffleMapStage 2 (MapPartitionsRDD[13] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:35:10 INFO MemoryStore: Block broadcast_5 stored as values in memory (estimated size 43.7 KiB, free
   433.8 MiB)
26/01/04 14:35:10 INFO MemoryStore: Block broadcast_5_piece0 stored as bytes in memory (estimated size 19.6
```

```
  KiB, free 433.8 MiB)
26/01/04 14:35:10 INFO SparkContext: Created broadcast 5 from broadcast at DAGScheduler.scala:1701
26/01/04 14:35:10 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 2 (MapPartitionsRDD[13] at
   $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:35:10 INFO TaskSchedulerImpl: Adding task set 2.0 with 1 tasks resource profile 0
26/01/04 14:35:10 INFO TaskSetManager: Starting task 0.0 in stage 2.0 (TID 2) (172.18.32.203,executor 0,
  partition 0, ANY, 10239 bytes)
26/01/04 14:35:27 INFO TaskSetManager: Finished task 0.0 in stage 2.0 (TID 2) in 16648 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:35:27 INFO TaskSchedulerImpl: Removed TaskSet 2.0 whose tasks have all completed, from pool
26/01/04 14:35:27 INFO DAGScheduler: ShuffleMapStage 2 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 16715 ms
26/01/04 14:35:27 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:35:27 INFO DAGScheduler: running: HashSet()
26/01/04 14:35:27 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:35:27 INFO DAGScheduler: failed: HashSet()
26/01/04 14:35:27 INFO ShufflePartitionsUtil: For shuffle(0, advisory target size: 67108864, actual target size
   1048576, minimum partition size: 1048576
26/01/04 14:35:27 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twolevel.enabled is set to true, but
  current version of codegened fast hashmap does not support this aggregate.
26/01/04 14:35:27 INFO CodeGenerator: Code generated in 182.888938 ms
26/01/04 14:35:28 INFO SparkContext: Starting job: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:35:28 INFO DAGScheduler: Got job 3 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:35:28 INFO DAGScheduler: Final stage: ResultStage 4 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:35:28 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 3)
26/01/04 14:35:28 INFO DAGScheduler: Missing parents: List()
26/01/04 14:35:28 INFO DAGScheduler: Missing parents found for ResultStage 4: List()
26/01/04 14:35:28 INFO DAGScheduler: Submitting ResultStage 4 (MapPartitionsRDD[16] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:35:28 INFO MemoryStore: Block broadcast_6 stored as values in memory (estimated size 49.9 KiB, free
   433.8 MiB)
26/01/04 14:35:28 INFO MemoryStore: Block broadcast_6_piece0 stored as bytes in memory (estimated size 22.0
  KiB, free 433.7 MiB)
26/01/04 14:35:28 INFO SparkContext: Created broadcast 6 from broadcast at DAGScheduler.scala:1701
26/01/04 14:35:28 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 4 (MapPartitionsRDD[16] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:35:28 INFO TaskSchedulerImpl: Adding task set 4.0 with 1 tasks resource profile 0
26/01/04 14:35:28 INFO TaskSetManager: Starting task 0.0 in stage 4.0 (TID 3) (172.18.32.203,executor 0,
  partition 0, NODE_LOCAL, 9635 bytes)
26/01/04 14:35:29 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 0 to
  172.18.32.203:50888
26/01/04 14:35:29 INFO TaskSetManager: Finished task 0.0 in stage 4.0 (TID 3) in 1761 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:35:29 INFO TaskSchedulerImpl: Removed TaskSet 4.0 whose tasks have all completed, from pool
26/01/04 14:35:29 INFO DAGScheduler: ResultStage 4 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 1824 ms
26/01/04 14:35:29 INFO DAGScheduler: Job 3 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:35:29 INFO TaskSchedulerImpl: Canceling stage 4
26/01/04 14:35:29 INFO TaskSchedulerImpl: Killing all running tasks in stage 4: Stage finished
26/01/04 14:35:29 INFO DAGScheduler: Job 3 finished: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768, took 1889.426175 ms
26/01/04 14:35:30 INFO CodeGenerator: Code generated in 52.686856 ms
+------+--------------------+----------------+
|   Sex|Survival_Rate_Percent|Total_Passengers|
+------+--------------------+----------------+
|female|    74.20382165605095|             314|
|  male|   18.890814558058924|             577|
+------+--------------------+----------------+


--- Survival Rate by Ticket Class (Pclass) ---
26/01/04 14:35:30 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:35:30 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/01/04 14:35:31 INFO CodeGenerator: Code generated in 287.333839 ms
26/01/04 14:35:31 INFO MemoryStore: Block broadcast_7 stored as values in memory (estimated size 215.5 KiB,
  free 433.7 MiB)
26/01/04 14:35:31 INFO MemoryStore: Block broadcast_7_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.7 MiB)
26/01/04 14:35:31 INFO SparkContext: Created broadcast 7 from $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:35:31 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:35:31 INFO DAGScheduler: Registering RDD 20 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 1
26/01/04 14:35:31 INFO DAGScheduler: Got map stage job 4 ($anonfun$withThreadLocalCaptured$2 at
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  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:35:31 INFO DAGScheduler: Final stage: ShuffleMapStage 5 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:35:31 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:35:31 INFO DAGScheduler: Missing parents: List()
26/01/04 14:35:31 INFO DAGScheduler: Missing parents found for ShuffleMapStage 5: List()
26/01/04 14:35:31 INFO DAGScheduler: Submitting ShuffleMapStage 5 (MapPartitionsRDD[20] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:35:31 INFO MemoryStore: Block broadcast_8 stored as values in memory (estimated size 43.4 KiB, free
   433.6 MiB)
26/01/04 14:35:31 INFO MemoryStore: Block broadcast_8_piece0 stored as bytes in memory (estimated size 19.5
  KiB, free 433.6 MiB)
26/01/04 14:35:31 INFO SparkContext: Created broadcast 8 from broadcast at DAGScheduler.scala:1701
26/01/04 14:35:31 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 5 (MapPartitionsRDD[20] at
   $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:35:31 INFO TaskSchedulerImpl: Adding task set 5.0 with 1 tasks resource profile 0
26/01/04 14:35:31 INFO TaskSetManager: Starting task 0.0 in stage 5.0 (TID 4) (172.18.32.200,executor 1,
  partition 0, ANY, 10239 bytes)
26/01/04 14:35:33 INFO TaskSetManager: Finished task 0.0 in stage 5.0 (TID 4) in 2261 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:35:33 INFO TaskSchedulerImpl: Removed TaskSet 5.0 whose tasks have all completed, from pool
26/01/04 14:35:33 INFO DAGScheduler: ShuffleMapStage 5 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 2296 ms
26/01/04 14:35:33 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:35:33 INFO DAGScheduler: running: HashSet()
26/01/04 14:35:33 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:35:33 INFO DAGScheduler: failed: HashSet()
26/01/04 14:35:33 INFO ShufflePartitionsUtil: For shuffle(1, advisory target size: 67108864, actual target size
   1048576, minimum partition size: 1048576
26/01/04 14:35:33 INFO CodeGenerator: Code generated in 68.668735 ms
26/01/04 14:35:33 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twolevel.enabled is set to true, but
  current version of codegened fast hashmap does not support this aggregate.
26/01/04 14:35:33 INFO CodeGenerator: Code generated in 154.795676 ms
26/01/04 14:35:34 INFO SparkContext: Starting job: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:35:34 INFO DAGScheduler: Got job 5 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:35:34 INFO DAGScheduler: Final stage: ResultStage 7 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:35:34 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 6)
26/01/04 14:35:34 INFO DAGScheduler: Missing parents: List()
26/01/04 14:35:34 INFO DAGScheduler: Missing parents found for ResultStage 7: List()
26/01/04 14:35:34 INFO DAGScheduler: Submitting ResultStage 7 (MapPartitionsRDD[24] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:35:34 INFO MemoryStore: Block broadcast_9 stored as values in memory (estimated size 50.6 KiB, free
   433.5 MiB)
26/01/04 14:35:34 INFO MemoryStore: Block broadcast_9_piece0 stored as bytes in memory (estimated size 22.3
  KiB, free 433.5 MiB)
26/01/04 14:35:34 INFO SparkContext: Created broadcast 9 from broadcast at DAGScheduler.scala:1701
26/01/04 14:35:34 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 7 (MapPartitionsRDD[24] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:35:34 INFO TaskSchedulerImpl: Adding task set 7.0 with 1 tasks resource profile 0
26/01/04 14:35:34 INFO TaskSetManager: Starting task 0.0 in stage 7.0 (TID 5) (172.18.32.200,executor 1,
  partition 0, NODE_LOCAL, 9635 bytes)
26/01/04 14:35:34 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 1 to
  172.18.32.200:56666
26/01/04 14:35:35 INFO TaskSetManager: Finished task 0.0 in stage 7.0 (TID 5) in 947 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:35:35 INFO TaskSchedulerImpl: Removed TaskSet 7.0 whose tasks have all completed, from pool
26/01/04 14:35:35 INFO DAGScheduler: ResultStage 7 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 992 ms
26/01/04 14:35:35 INFO DAGScheduler: Job 5 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:35:35 INFO TaskSchedulerImpl: Canceling stage 7
26/01/04 14:35:35 INFO TaskSchedulerImpl: Killing all running tasks in stage 7: Stage finished
26/01/04 14:35:35 INFO DAGScheduler: Job 5 finished: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768, took 1033.98681 ms
26/01/04 14:35:35 INFO CodeGenerator: Code generated in 59.358524 ms
+------+--------------------+----------------+
|Pclass|Survival_Rate_Percent|Total_Passengers|
+------+--------------------+----------------+
|     1|    62.96296296296296|             216|
|     2|    47.28260869565217|             184|
|     3|   24.236252545824847|             491|
+------+--------------------+----------------+


--- Average Age: Survivors vs Non-Survivors ---
26/01/04 14:35:35 INFO FileSourceStrategy: Pushed Filters:
```

```
26/01/04 14:35:35 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/01/04 14:35:35 INFO CodeGenerator: Code generated in 144.035232 ms
26/01/04 14:35:35 INFO MemoryStore: Block broadcast_10 stored as values in memory (estimated size 215.5 KiB,
  free 433.3 MiB)
26/01/04 14:35:35 INFO MemoryStore: Block broadcast_10_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.3 MiB)
26/01/04 14:35:35 INFO SparkContext: Created broadcast 10 from $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:35:35 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:35:35 INFO DAGScheduler: Registering RDD 28 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 2
26/01/04 14:35:35 INFO DAGScheduler: Got map stage job 6 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:35:35 INFO DAGScheduler: Final stage: ShuffleMapStage 8 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:35:35 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:35:35 INFO DAGScheduler: Missing parents: List()
26/01/04 14:35:35 INFO DAGScheduler: Missing parents found for ShuffleMapStage 8: List()
26/01/04 14:35:35 INFO DAGScheduler: Submitting ShuffleMapStage 8 (MapPartitionsRDD[28] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:35:35 INFO MemoryStore: Block broadcast_11 stored as values in memory (estimated size 41.5 KiB,
  free 433.2 MiB)
26/01/04 14:35:35 INFO MemoryStore: Block broadcast_11_piece0 stored as bytes in memory (estimated size 18.8
  KiB, free 433.2 MiB)
26/01/04 14:35:35 INFO SparkContext: Created broadcast 11 from broadcast at DAGScheduler.scala:1701
26/01/04 14:35:35 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 8 (MapPartitionsRDD[28] at
   $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:35:35 INFO TaskSchedulerImpl: Adding task set 8.0 with 1 tasks resource profile 0
26/01/04 14:35:35 INFO TaskSetManager: Starting task 0.0 in stage 8.0 (TID 6) (172.18.32.203,executor 0,
  partition 0, ANY, 10239 bytes)
26/01/04 14:35:37 INFO TaskSetManager: Finished task 0.0 in stage 8.0 (TID 6) in 1636 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:35:37 INFO TaskSchedulerImpl: Removed TaskSet 8.0 whose tasks have all completed, from pool
26/01/04 14:35:37 INFO DAGScheduler: ShuffleMapStage 8 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 1662 ms
26/01/04 14:35:37 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:35:37 INFO DAGScheduler: running: HashSet()
26/01/04 14:35:37 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:35:37 INFO DAGScheduler: failed: HashSet()
26/01/04 14:35:37 INFO ShufflePartitionsUtil: For shuffle(2, advisory target size: 67108864, actual target size
   1048576, minimum partition size: 1048576
26/01/04 14:35:37 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twolevel.enabled is set to true, but
  current version of codegened fast hashmap does not support this aggregate.
26/01/04 14:35:37 INFO CodeGenerator: Code generated in 96.034742 ms
26/01/04 14:35:37 INFO SparkContext: Starting job: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:35:37 INFO DAGScheduler: Got job 7 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:35:37 INFO DAGScheduler: Final stage: ResultStage 10 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:35:37 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 9)
26/01/04 14:35:37 INFO DAGScheduler: Missing parents: List()
26/01/04 14:35:37 INFO DAGScheduler: Missing parents found for ResultStage 10: List()
26/01/04 14:35:37 INFO DAGScheduler: Submitting ResultStage 10 (MapPartitionsRDD[31] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:35:37 INFO MemoryStore: Block broadcast_12 stored as values in memory (estimated size 48.7 KiB,
  free 433.2 MiB)
26/01/04 14:35:37 INFO MemoryStore: Block broadcast_12_piece0 stored as bytes in memory (estimated size 21.3
  KiB, free 433.1 MiB)
26/01/04 14:35:37 INFO SparkContext: Created broadcast 12 from broadcast at DAGScheduler.scala:1701
26/01/04 14:35:37 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 10 (MapPartitionsRDD[31] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:35:37 INFO TaskSchedulerImpl: Adding task set 10.0 with 1 tasks resource profile 0
26/01/04 14:35:37 INFO TaskSetManager: Starting task 0.0 in stage 10.0 (TID 7) (172.18.32.203,executor 0,
  partition 0, NODE_LOCAL, 9635 bytes)
26/01/04 14:35:37 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 2 to
  172.18.32.203:50888
26/01/04 14:35:38 INFO TaskSetManager: Finished task 0.0 in stage 10.0 (TID 7) in 606 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:35:38 INFO TaskSchedulerImpl: Removed TaskSet 10.0 whose tasks have all completed, from pool
26/01/04 14:35:38 INFO DAGScheduler: ResultStage 10 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 640 ms
26/01/04 14:35:38 INFO DAGScheduler: Job 7 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:35:38 INFO TaskSchedulerImpl: Canceling stage 10
26/01/04 14:35:38 INFO TaskSchedulerImpl: Killing all running tasks in stage 10: Stage finished
26/01/04 14:35:38 INFO DAGScheduler: Job 7 finished: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768, took 660.544159 ms
```

```
+--------+------------------+------------------+
|Survived|       Average_Age|Count_With_Age_Data|
+--------+------------------+------------------+
|       1|28.343689655172415|               290|
|       0| 30.62617924528302|               424|
+--------+------------------+------------------+

26/01/04 14:35:39 INFO SparkContext: Invoking stop() from shutdown hook
26/01/04 14:35:39 INFO SparkContext: SparkContext is stopping with exitCode 0 from run at Executors.java:572.
26/01/04 14:35:39 INFO SparkUI: Stopped Spark web UI at http://raspberrypi-dml0:4040
26/01/04 14:35:39 INFO StandaloneSchedulerBackend: Shutting down all executors
26/01/04 14:35:39 INFO StandaloneSchedulerBackend$StandaloneDriverEndpoint: Asking each executor to shut down
26/01/04 14:35:39 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/01/04 14:35:39 INFO MemoryStore: MemoryStore cleared
26/01/04 14:35:39 INFO BlockManager: BlockManager stopped
26/01/04 14:35:39 INFO BlockManagerMaster: BlockManagerMaster stopped
26/01/04 14:35:39 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator
  stopped!
26/01/04 14:35:39 INFO SparkContext: Successfully stopped SparkContext (Uptime: 85187 ms)
karimi@raspberrypi-dml0:~/CA3 $
```

3, 4

After adding this code to my python code:

```
# ========================================
# PART 3.3 & 3.4: Machine Learning Pipeline
# ========================================
from pyspark.ml.feature import StringIndexer, VectorAssembler
from pyspark.ml.classification import LogisticRegression
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
import os

print("\n--- Starting Machine Learning Pipeline ---")

# 1. Prepare Data (Handle Nulls)
# We drop rows with missing values in the columns we care about
required_columns = ["Pclass", "Sex", "Age", "Fare", "Embarked", "Survived"]
data = df.select(required_columns).dropna()

# 2. Convert Categorical Columns to Numbers (StringIndexer)
# Sex: male/female -> 0/1
sex_indexer = StringIndexer(inputCol="Sex", outputCol="SexIndex")
data = sex_indexer.fit(data).transform(data)

# Embarked: S/C/Q -> 0/1/2
embarked_indexer = StringIndexer(inputCol="Embarked", outputCol="EmbarkedIndex")
data = embarked_indexer.fit(data).transform(data)

# 3. Assemble Features
# Combine all input columns into a single vector column named "features"
feature_cols = ["Pclass", "SexIndex", "Age", "Fare", "EmbarkedIndex"]
assembler = VectorAssembler(inputCols=feature_cols, outputCol="features")
final_data = assembler.transform(data)

# 4. Split Data (80% Train, 20% Test)
train_data, test_data = final_data.randomSplit([0.8, 0.2], seed=42)

print(f"Training Dataset Count: {train_data.count()}")
print(f"Test Dataset Count: {test_data.count()}")

# 5. Train Logistic Regression Model
lr = LogisticRegression(featuresCol="features", labelCol="Survived")
lr_model = lr.fit(train_data)

# 6. Make Predictions
predictions = lr_model.transform(test_data)

# 7. Evaluate Accuracy
evaluator = MulticlassClassificationEvaluator(
    labelCol="Survived", predictionCol="prediction", metricName="accuracy"
)
accuracy = evaluator.evaluate(predictions)
print(f"\nModel Accuracy on Test Data: {accuracy:.4f}")
```

```
# ========================================
# PART 3.4: Save Result to HDFS
# ========================================

# Step A: Write to a LOCAL file first
local_filename = "accuracy.txt"
with open(local_filename, "w") as f:
    f.write(f"Accuracy: {accuracy:.4f}")

print(f"Accuracy saved locally to {local_filename}")

# Step B: Upload this local file to your HDFS folder
# We use os.system to run the shell command from Python
# Note: Using the path /karimi/ based on your previous output
hdfs_dest = "/karimi/accuracy.txt"
upload_cmd = f"hdfs dfs -put -f {local_filename} {hdfs_dest}"

print(f"Uploading to HDFS: {upload_cmd}")
exit_code = os.system(upload_cmd)

if exit_code == 0:
    print("Successfully uploaded accuracy.txt to HDFS!")
else:
    print("Error uploading to HDFS. Please check permissions.")
```

And this is the whole result:

```
karimi@raspberrypi-dml0:~/CA3 $ nano titanic_hdfs.py
karimi@raspberrypi-dml0:~/CA3 $ cat titanic_hdfs.py
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, avg, count, when

# 1. Initialize Spark Session
spark = SparkSession.builder \
    .appName("Titanic_Analysis_Karimi") \
    .getOrCreate()

# 2. Load Data from HDFS
# Based on your terminal output, the file is at /karimi/Titanic-Dataset.csv
hdfs_path = "hdfs://172.18.32.200:9000/karimi/Titanic-Dataset.csv"

print(f"Loading data from: {hdfs_path} ...")
df = spark.read.csv(hdfs_path, header=True, inferSchema=True)

# 3. Calculate Survival Rate by Sex (Question 3.2 Part 1)
print("\n--- Survival Rate by Sex ---")
# Group by Sex, calculate average of 'Survived' (0 or 1)
# Since Survived is 0 or 1, the average * 100 gives the percentage.
sex_survival = df.groupBy("Sex").agg(
    (avg("Survived") * 100).alias("Survival_Rate_Percent"),
    count("PassengerId").alias("Total_Passengers")
)
sex_survival.show()

# 4. Calculate Survival Rate by Pclass (Question 3.2 Part 2)
print("\n--- Survival Rate by Ticket Class (Pclass) ---")
class_survival = df.groupBy("Pclass").agg(
    (avg("Survived") * 100).alias("Survival_Rate_Percent"),
    count("PassengerId").alias("Total_Passengers")
).orderBy("Pclass")
class_survival.show()

# 5. Average Age for Survivors vs Non-Survivors (Question 3.2 Part 3)
print("\n--- Average Age: Survivors vs Non-Survivors ---")
# 0 = No (Deceased), 1 = Yes (Survived)
age_analysis = df.groupBy("Survived").agg(
    avg("Age").alias("Average_Age"),
    count("Age").alias("Count_With_Age_Data") # Just to see how many had valid ages
)
age_analysis.show()

# ========================================
# PART 3.3 & 3.4: Machine Learning Pipeline
# ========================================
from pyspark.ml.feature import StringIndexer, VectorAssembler
from pyspark.ml.classification import LogisticRegression
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
import os
```

```
print("\n--- Starting Machine Learning Pipeline ---")

# 1. Prepare Data (Handle Nulls)
# We drop rows with missing values in the columns we care about
required_columns = ["Pclass", "Sex", "Age", "Fare", "Embarked", "Survived"]
data = df.select(required_columns).dropna()

# 2. Convert Categorical Columns to Numbers (StringIndexer)
# Sex: male/female -> 0/1
sex_indexer = StringIndexer(inputCol="Sex", outputCol="SexIndex")
data = sex_indexer.fit(data).transform(data)

# Embarked: S/C/Q -> 0/1/2
embarked_indexer = StringIndexer(inputCol="Embarked", outputCol="EmbarkedIndex")
data = embarked_indexer.fit(data).transform(data)

# 3. Assemble Features
# Combine all input columns into a single vector column named "features"
feature_cols = ["Pclass", "SexIndex", "Age", "Fare", "EmbarkedIndex"]
assembler = VectorAssembler(inputCols=feature_cols, outputCol="features")
final_data = assembler.transform(data)

# 4. Split Data (80% Train, 20% Test)
train_data, test_data = final_data.randomSplit([0.8, 0.2], seed=42)

print(f"Training Dataset Count: {train_data.count()}")
print(f"Test Dataset Count: {test_data.count()}")

# 5. Train Logistic Regression Model
lr = LogisticRegression(featuresCol="features", labelCol="Survived")
lr_model = lr.fit(train_data)

# 6. Make Predictions
predictions = lr_model.transform(test_data)

# 7. Evaluate Accuracy
evaluator = MulticlassClassificationEvaluator(
    labelCol="Survived", predictionCol="prediction", metricName="accuracy"
)
accuracy = evaluator.evaluate(predictions)
print(f"\nModel Accuracy on Test Data: {accuracy:.4f}")

# ==========================================
# PART 3.4: Save Result to HDFS
# ==========================================

# Step A: Write to a LOCAL file first
local_filename = "accuracy.txt"
with open(local_filename, "w") as f:
    f.write(f"Accuracy: {accuracy:.4f}")

print(f"Accuracy saved locally to {local_filename}")

# Step B: Upload this local file to your HDFS folder
# We use os.system to run the shell command from Python
# Note: Using the path /karimi/ based on your previous output
hdfs_dest = "/karimi/accuracy.txt"
upload_cmd = f"hdfs dfs -put -f {local_filename} {hdfs_dest}"

print(f"Uploading to HDFS: {upload_cmd}")
exit_code = os.system(upload_cmd)

if exit_code == 0:
    print("Successfully uploaded accuracy.txt to HDFS!")
else:
    print("Error uploading to HDFS. Please check permissions.")
karimi@raspberrypi-dml0:~/CA3 $ spark-submit titanic_hdfs.py
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/01/04 14:40:27 INFO SparkContext: Running Spark version 4.1.0
26/01/04 14:40:27 INFO SparkContext: OS info Linux, 6.6.42-v8+, aarch64
26/01/04 14:40:27 INFO SparkContext: Java version 21.0.9+7-LTS-338
26/01/04 14:40:28 INFO ResourceUtils: ==============================================================
26/01/04 14:40:28 INFO ResourceUtils: No custom resources configured for spark.driver.
26/01/04 14:40:28 INFO ResourceUtils: ==============================================================
26/01/04 14:40:28 INFO SparkContext: Submitted application: Titanic_Analysis_Karimi
26/01/04 14:40:28 INFO SecurityManager: Changing view acls to: hadoop,dousti,karimi
26/01/04 14:40:28 INFO SecurityManager: Changing modify acls to: hadoop,dousti,karimi
26/01/04 14:40:28 INFO SecurityManager: Changing view acls groups to: hadoop,dousti,karimi
26/01/04 14:40:28 INFO SecurityManager: Changing modify acls groups to: hadoop,dousti,karimi
26/01/04 14:40:28 INFO SecurityManager: SecurityManager: authentication disabled; ui acls enabled; users with
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  view permissions: hadoop, dousti, karimi groups with view permissions: EMPTY; users with modify permissions:
  hadoop, dousti, karimi; groups with modify permissions: EMPTY; RPC SSL disabled
26/01/04 14:40:29 INFO Utils: Successfully started service 'sparkDriver' on port 41345.
26/01/04 14:40:29 INFO SparkEnv: Registering MapOutputTracker
26/01/04 14:40:29 INFO SparkEnv: Registering BlockManagerMaster
26/01/04 14:40:29 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for
  getting topology information
26/01/04 14:40:29 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/01/04 14:40:29 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/01/04 14:40:30 INFO DiskBlockManager: Created local directory at
  /tmp/blockmgr-acd42f25-dbc0-4040-ba44-efe6657702b4
26/01/04 14:40:30 INFO SparkEnv: Registering OutputCommitCoordinator
26/01/04 14:40:30 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI
26/01/04 14:40:30 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/01/04 14:40:30 INFO ServerInfo: Adding filter to /: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:31 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name:
   cores, amount: 2, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap ->
  name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/01/04 14:40:31 INFO ResourceProfile: Limiting resource is cpus at 2 tasks per executor
26/01/04 14:40:31 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/01/04 14:40:31 INFO SecurityManager: Changing view acls to: hadoop,dousti,karimi
26/01/04 14:40:31 INFO SecurityManager: Changing modify acls to: hadoop,dousti,karimi
26/01/04 14:40:31 INFO SecurityManager: Changing view acls groups to: hadoop,dousti,karimi
26/01/04 14:40:31 INFO SecurityManager: Changing modify acls groups to: hadoop,dousti,karimi
26/01/04 14:40:31 INFO SecurityManager: SecurityManager: authentication disabled; ui acls enabled; users with
  view permissions: hadoop, dousti, karimi groups with view permissions: EMPTY; users with modify permissions:
  hadoop, dousti, karimi; groups with modify permissions: EMPTY; RPC SSL disabled
26/01/04 14:40:32 INFO StandaloneAppClient$ClientEndpoint: Connecting to master spark://172.18.32.200:7077...
26/01/04 14:40:32 INFO TransportClientFactory: Successfully created connection to /172.18.32.200:7077 after 172
   ms (0 ms spent in bootstraps)
26/01/04 14:40:32 INFO StandaloneSchedulerBackend: Connected to Spark cluster with app ID
  app-20260104144032-0033
26/01/04 14:40:32 INFO StandaloneAppClient$ClientEndpoint: Executor added: app-20260104144032-0033/0 on
  worker-20260104001835-172.18.32.203-41527 (172.18.32.203:41527) with 2 core(s)
26/01/04 14:40:32 INFO StandaloneSchedulerBackend: Granted executor ID app-20260104144032-0033/0 on hostPort
  172.18.32.203:41527 with 2 core(s), 1024.0 MiB RAM
26/01/04 14:40:32 INFO StandaloneAppClient$ClientEndpoint: Executor added: app-20260104144032-0033/1 on
  worker-20260104001832-172.18.32.200-35083 (172.18.32.200:35083) with 2 core(s)
26/01/04 14:40:32 INFO StandaloneSchedulerBackend: Granted executor ID app-20260104144032-0033/1 on hostPort
  172.18.32.200:35083 with 2 core(s), 1024.0 MiB RAM
26/01/04 14:40:32 INFO Utils: Successfully started service
  'org.apache.spark.network.netty.NettyBlockTransferService' on port 45297.
26/01/04 14:40:32 INFO NettyBlockTransferService: Server created on raspberrypi-dml0:45297
26/01/04 14:40:32 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block
  replication policy
26/01/04 14:40:33 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, raspberrypi-dml0,
  45297, None)
26/01/04 14:40:33 INFO StandaloneAppClient$ClientEndpoint: Executor updated: app-20260104144032-0033/1 is now
  RUNNING
26/01/04 14:40:33 INFO BlockManagerMasterEndpoint: Registering block manager raspberrypi-dml0:45297 with 434.4
  MiB RAM, BlockManagerId(driver, raspberrypi-dml0, 45297, None)
26/01/04 14:40:33 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, raspberrypi-dml0,
  45297, None)
26/01/04 14:40:33 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, raspberrypi-dml0, 45297,
  None)
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /jobs: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /jobs/json: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /jobs/job: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /jobs/job/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO StandaloneAppClient$ClientEndpoint: Executor updated: app-20260104144032-0033/0 is now
  RUNNING
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /stages: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /stages/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /stages/stage:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /stages/stage/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /stages/pool:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /stages/pool/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /stages/taskThreadDump:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /stages/taskThreadDump/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /storage: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /storage/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /storage/rdd:
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  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:33 INFO ServerInfo: Adding filter to /storage/rdd/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /environment:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /environment/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /executors: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /executors/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /executors/threadDump:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /executors/threadDump/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /executors/heapHistogram:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /executors/heapHistogram/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /static: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /api: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /metrics: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /jobs/job/kill:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /stages/stage/kill:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO ServerInfo: Adding filter to /metrics/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:34 INFO StandaloneSchedulerBackend: SchedulerBackend is ready for scheduling beginning after
  reached minRegisteredResourcesRatio: 0.0
26/01/04 14:40:47 INFO StandaloneSchedulerBackend$StandaloneDriverEndpoint: Registered executor
  NettyRpcEndpointRef(spark-client://Executor) (172.18.32.200:56416) with ID 1, ResourceProfileId 0
26/01/04 14:40:48 INFO BlockManagerMasterEndpoint: Registering block manager 172.18.32.200:34187 with 434.4 MiB
   RAM, BlockManagerId(1, 172.18.32.200, 34187, None)
Loading data from: hdfs://172.18.32.200:9000/karimi/Titanic-Dataset.csv ...
26/01/04 14:40:49 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of
  spark.sql.warehouse.dir.
26/01/04 14:40:49 INFO SharedState: Warehouse path is 'file:/home/dmls/karimi/CA3/spark-warehouse'.
26/01/04 14:40:49 INFO ServerInfo: Adding filter to /SQL: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:49 INFO ServerInfo: Adding filter to /SQL/json: com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:49 INFO ServerInfo: Adding filter to /SQL/execution:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:49 INFO ServerInfo: Adding filter to /SQL/execution/json:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:49 INFO ServerInfo: Adding filter to /static/sql:
  com.neolitec.examples.BasicAuthenticationFilter
26/01/04 14:40:52 INFO StandaloneSchedulerBackend$StandaloneDriverEndpoint: Registered executor
  NettyRpcEndpointRef(spark-client://Executor) (172.18.32.203:57392) with ID 0, ResourceProfileId 0
26/01/04 14:40:53 INFO InMemoryFileIndex: It took 251 ms to list leaf files for 1 paths.
26/01/04 14:40:53 INFO InMemoryFileIndex: It took 17 ms to list leaf files for 1 paths.
26/01/04 14:40:54 INFO BlockManagerMasterEndpoint: Registering block manager 172.18.32.203:33925 with 434.4 MiB
   RAM, BlockManagerId(0, 172.18.32.203, 33925, None)
26/01/04 14:40:56 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:40:56 INFO FileSourceStrategy: Post-Scan Filters: Set((length(trim(value#0, None)) > 0))
26/01/04 14:40:58 INFO CodeGenerator: Code generated in 838.992411 ms
26/01/04 14:40:58 INFO MemoryStore: MemoryStore started with capacity 434.4 MiB
26/01/04 14:40:58 INFO MemoryStore: Block broadcast_0 stored as values in memory (estimated size 215.6 KiB,
  free 434.2 MiB)
26/01/04 14:40:58 INFO MemoryStore: Block broadcast_0_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 434.2 MiB)
26/01/04 14:40:58 INFO SparkContext: Created broadcast 0 from csv at NativeMethodAccessorImpl.java:0
26/01/04 14:40:58 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:40:59 INFO SparkContext: Starting job: csv at NativeMethodAccessorImpl.java:0
26/01/04 14:40:59 INFO DAGScheduler: Got job 0 (csv at NativeMethodAccessorImpl.java:0) with 1 output
  partitions
26/01/04 14:40:59 INFO DAGScheduler: Final stage: ResultStage 0 (csv at NativeMethodAccessorImpl.java:0)
26/01/04 14:40:59 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:40:59 INFO DAGScheduler: Missing parents: List()
26/01/04 14:40:59 INFO DAGScheduler: Missing parents found for ResultStage 0: List()
26/01/04 14:40:59 INFO DAGScheduler: Submitting ResultStage 0 (MapPartitionsRDD[3] at csv at
  NativeMethodAccessorImpl.java:0), which has no missing parents
26/01/04 14:40:59 INFO MemoryStore: Block broadcast_1 stored as values in memory (estimated size 14.3 KiB, free
   434.1 MiB)
26/01/04 14:40:59 INFO MemoryStore: Block broadcast_1_piece0 stored as bytes in memory (estimated size 6.8 KiB,
   free 434.1 MiB)
26/01/04 14:40:59 INFO SparkContext: Created broadcast 1 from broadcast at DAGScheduler.scala:1701
26/01/04 14:40:59 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 0 (MapPartitionsRDD[3] at csv
  at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
26/01/04 14:40:59 INFO TaskSchedulerImpl: Adding task set 0.0 with 1 tasks resource profile 0
26/01/04 14:40:59 INFO TaskSetManager: Starting task 0.0 in stage 0.0 (TID 0) (172.18.32.203,executor 0,
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  partition 0, ANY, 10250 bytes)
26/01/04 14:41:13 INFO TaskSetManager: Finished task 0.0 in stage 0.0 (TID 0) in 13978 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:41:13 INFO TaskSchedulerImpl: Removed TaskSet 0.0 whose tasks have all completed, from pool
26/01/04 14:41:13 INFO DAGScheduler: ResultStage 0 (csv at NativeMethodAccessorImpl.java:0) finished in 14390
  ms
26/01/04 14:41:13 INFO DAGScheduler: Job 0 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:41:13 INFO TaskSchedulerImpl: Canceling stage 0
26/01/04 14:41:13 INFO TaskSchedulerImpl: Killing all running tasks in stage 0: Stage finished
26/01/04 14:41:13 INFO DAGScheduler: Job 0 finished: csv at NativeMethodAccessorImpl.java:0, took 14568.116074
  ms
26/01/04 14:41:13 INFO CodeGenerator: Code generated in 47.270497 ms
26/01/04 14:41:13 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:41:13 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/01/04 14:41:14 INFO MemoryStore: Block broadcast_2 stored as values in memory (estimated size 215.6 KiB,
  free 433.9 MiB)
26/01/04 14:41:14 INFO MemoryStore: Block broadcast_2_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.9 MiB)
26/01/04 14:41:14 INFO SparkContext: Created broadcast 2 from csv at NativeMethodAccessorImpl.java:0
26/01/04 14:41:14 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:41:14 INFO SparkContext: Starting job: csv at NativeMethodAccessorImpl.java:0
26/01/04 14:41:14 INFO DAGScheduler: Got job 1 (csv at NativeMethodAccessorImpl.java:0) with 1 output
  partitions
26/01/04 14:41:14 INFO DAGScheduler: Final stage: ResultStage 1 (csv at NativeMethodAccessorImpl.java:0)
26/01/04 14:41:14 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:41:14 INFO DAGScheduler: Missing parents: List()
26/01/04 14:41:14 INFO DAGScheduler: Missing parents found for ResultStage 1: List()
26/01/04 14:41:14 INFO DAGScheduler: Submitting ResultStage 1 (MapPartitionsRDD[9] at csv at
  NativeMethodAccessorImpl.java:0), which has no missing parents
26/01/04 14:41:14 INFO MemoryStore: Block broadcast_3 stored as values in memory (estimated size 23.4 KiB, free
   433.9 MiB)
26/01/04 14:41:14 INFO MemoryStore: Block broadcast_3_piece0 stored as bytes in memory (estimated size 10.7
  KiB, free 433.8 MiB)
26/01/04 14:41:14 INFO SparkContext: Created broadcast 3 from broadcast at DAGScheduler.scala:1701
26/01/04 14:41:14 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 1 (MapPartitionsRDD[9] at csv
  at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
26/01/04 14:41:14 INFO TaskSchedulerImpl: Adding task set 1.0 with 1 tasks resource profile 0
26/01/04 14:41:14 INFO TaskSetManager: Starting task 0.0 in stage 1.0 (TID 1) (172.18.32.203,executor 0,
  partition 0, ANY, 10250 bytes)
26/01/04 14:41:24 INFO TaskSetManager: Finished task 0.0 in stage 1.0 (TID 1) in 9987 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:41:24 INFO TaskSchedulerImpl: Removed TaskSet 1.0 whose tasks have all completed, from pool
26/01/04 14:41:24 INFO DAGScheduler: ResultStage 1 (csv at NativeMethodAccessorImpl.java:0) finished in 10106
  ms
26/01/04 14:41:24 INFO DAGScheduler: Job 1 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:41:24 INFO TaskSchedulerImpl: Canceling stage 1
26/01/04 14:41:24 INFO TaskSchedulerImpl: Killing all running tasks in stage 1: Stage finished
26/01/04 14:41:24 INFO DAGScheduler: Job 1 finished: csv at NativeMethodAccessorImpl.java:0, took 10148.774332
  ms

--- Survival Rate by Sex ---
26/01/04 14:41:29 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:41:29 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/01/04 14:41:30 INFO CodeGenerator: Code generated in 307.976912 ms
26/01/04 14:41:30 INFO MemoryStore: Block broadcast_4 stored as values in memory (estimated size 215.5 KiB,
  free 434.2 MiB)
26/01/04 14:41:30 INFO MemoryStore: Block broadcast_4_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 434.2 MiB)
26/01/04 14:41:30 INFO SparkContext: Created broadcast 4 from $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:41:30 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:41:30 INFO DAGScheduler: Registering RDD 13 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 0
26/01/04 14:41:30 INFO DAGScheduler: Got map stage job 2 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:41:30 INFO DAGScheduler: Final stage: ShuffleMapStage 2 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:41:30 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:41:30 INFO DAGScheduler: Missing parents: List()
26/01/04 14:41:30 INFO DAGScheduler: Missing parents found for ShuffleMapStage 2: List()
26/01/04 14:41:30 INFO DAGScheduler: Submitting ShuffleMapStage 2 (MapPartitionsRDD[13] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:41:30 INFO MemoryStore: Block broadcast_5 stored as values in memory (estimated size 43.8 KiB, free
   434.1 MiB)
26/01/04 14:41:30 INFO MemoryStore: Block broadcast_5_piece0 stored as bytes in memory (estimated size 19.6
  KiB, free 434.1 MiB)
26/01/04 14:41:30 INFO SparkContext: Created broadcast 5 from broadcast at DAGScheduler.scala:1701
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26/01/04 14:41:30 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 2 (MapPartitionsRDD[13] at
   $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:41:30 INFO TaskSchedulerImpl: Adding task set 2.0 with 1 tasks resource profile 0
26/01/04 14:41:30 INFO TaskSetManager: Starting task 0.0 in stage 2.0 (TID 2) (172.18.32.203,executor 0,
  partition 0, ANY, 10239 bytes)
26/01/04 14:41:34 INFO TaskSetManager: Finished task 0.0 in stage 2.0 (TID 2) in 4132 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:41:34 INFO TaskSchedulerImpl: Removed TaskSet 2.0 whose tasks have all completed, from pool
26/01/04 14:41:34 INFO DAGScheduler: ShuffleMapStage 2 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 4218 ms
26/01/04 14:41:34 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:41:34 INFO DAGScheduler: running: HashSet()
26/01/04 14:41:34 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:41:34 INFO DAGScheduler: failed: HashSet()
26/01/04 14:41:34 INFO ShufflePartitionsUtil: For shuffle(0, advisory target size: 67108864, actual target size
   1048576, minimum partition size: 1048576
26/01/04 14:41:35 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twolevel.enabled is set to true, but
  current version of codegened fast hashmap does not support this aggregate.
26/01/04 14:41:35 INFO CodeGenerator: Code generated in 183.839293 ms
26/01/04 14:41:35 INFO SparkContext: Starting job: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:41:35 INFO DAGScheduler: Got job 3 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:41:35 INFO DAGScheduler: Final stage: ResultStage 4 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:41:35 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 3)
26/01/04 14:41:35 INFO DAGScheduler: Missing parents: List()
26/01/04 14:41:35 INFO DAGScheduler: Missing parents found for ResultStage 4: List()
26/01/04 14:41:35 INFO DAGScheduler: Submitting ResultStage 4 (MapPartitionsRDD[16] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:41:35 INFO MemoryStore: Block broadcast_6 stored as values in memory (estimated size 49.9 KiB, free
   434.0 MiB)
26/01/04 14:41:35 INFO MemoryStore: Block broadcast_6_piece0 stored as bytes in memory (estimated size 22.0
  KiB, free 434.0 MiB)
26/01/04 14:41:35 INFO SparkContext: Created broadcast 6 from broadcast at DAGScheduler.scala:1701
26/01/04 14:41:35 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 4 (MapPartitionsRDD[16] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:41:35 INFO TaskSchedulerImpl: Adding task set 4.0 with 1 tasks resource profile 0
26/01/04 14:41:35 INFO TaskSetManager: Starting task 0.0 in stage 4.0 (TID 3) (172.18.32.203,executor 0,
  partition 0, NODE_LOCAL, 9635 bytes)
26/01/04 14:41:36 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 0 to
  172.18.32.203:57392
26/01/04 14:41:37 INFO TaskSetManager: Finished task 0.0 in stage 4.0 (TID 3) in 1464 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:41:37 INFO TaskSchedulerImpl: Removed TaskSet 4.0 whose tasks have all completed, from pool
26/01/04 14:41:37 INFO DAGScheduler: ResultStage 4 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 1526 ms
26/01/04 14:41:37 INFO DAGScheduler: Job 3 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:41:37 INFO TaskSchedulerImpl: Canceling stage 4
26/01/04 14:41:37 INFO TaskSchedulerImpl: Killing all running tasks in stage 4: Stage finished
26/01/04 14:41:37 INFO DAGScheduler: Job 3 finished: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768, took 1588.419994 ms
26/01/04 14:41:37 INFO CodeGenerator: Code generated in 62.638621 ms
+------+--------------------+----------------+
|   Sex|Survival_Rate_Percent|Total_Passengers|
+------+--------------------+----------------+
|female|    74.20382165605095|             314|
|  male|   18.890814558058924|             577|
+------+--------------------+----------------+


--- Survival Rate by Ticket Class (Pclass) ---
26/01/04 14:41:37 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:41:37 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/01/04 14:41:38 INFO CodeGenerator: Code generated in 333.534002 ms
26/01/04 14:41:38 INFO MemoryStore: Block broadcast_7 stored as values in memory (estimated size 215.5 KiB,
  free 433.8 MiB)
26/01/04 14:41:38 INFO MemoryStore: Block broadcast_7_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.8 MiB)
26/01/04 14:41:38 INFO SparkContext: Created broadcast 7 from $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:41:38 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:41:38 INFO DAGScheduler: Registering RDD 20 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 1
26/01/04 14:41:38 INFO DAGScheduler: Got map stage job 4 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:41:38 INFO DAGScheduler: Final stage: ShuffleMapStage 5 ($anonfun$withThreadLocalCaptured$2 at
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  CompletableFuture.java:1768)
26/01/04 14:41:38 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:41:38 INFO DAGScheduler: Missing parents: List()
26/01/04 14:41:38 INFO DAGScheduler: Missing parents found for ShuffleMapStage 5: List()
26/01/04 14:41:38 INFO DAGScheduler: Submitting ShuffleMapStage 5 (MapPartitionsRDD[20] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:41:38 INFO MemoryStore: Block broadcast_8 stored as values in memory (estimated size 43.4 KiB, free
   434.0 MiB)
26/01/04 14:41:38 INFO MemoryStore: Block broadcast_8_piece0 stored as bytes in memory (estimated size 19.5
  KiB, free 434.0 MiB)
26/01/04 14:41:38 INFO SparkContext: Created broadcast 8 from broadcast at DAGScheduler.scala:1701
26/01/04 14:41:38 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 5 (MapPartitionsRDD[20] at
   $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:41:38 INFO TaskSchedulerImpl: Adding task set 5.0 with 1 tasks resource profile 0
26/01/04 14:41:38 INFO TaskSetManager: Starting task 0.0 in stage 5.0 (TID 4) (172.18.32.203,executor 0,
  partition 0, ANY, 10239 bytes)
26/01/04 14:41:39 INFO TaskSetManager: Finished task 0.0 in stage 5.0 (TID 4) in 1487 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:41:39 INFO TaskSchedulerImpl: Removed TaskSet 5.0 whose tasks have all completed, from pool
26/01/04 14:41:39 INFO DAGScheduler: ShuffleMapStage 5 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 1538 ms
26/01/04 14:41:39 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:41:39 INFO DAGScheduler: running: HashSet()
26/01/04 14:41:39 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:41:39 INFO DAGScheduler: failed: HashSet()
26/01/04 14:41:39 INFO ShufflePartitionsUtil: For shuffle(1, advisory target size: 67108864, actual target size
   1048576, minimum partition size: 1048576
26/01/04 14:41:40 INFO CodeGenerator: Code generated in 100.791832 ms
26/01/04 14:41:40 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twolevel.enabled is set to true, but
  current version of codegened fast hashmap does not support this aggregate.
26/01/04 14:41:40 INFO CodeGenerator: Code generated in 186.055781 ms
26/01/04 14:41:40 INFO SparkContext: Starting job: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:41:40 INFO DAGScheduler: Got job 5 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:41:40 INFO DAGScheduler: Final stage: ResultStage 7 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:41:40 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 6)
26/01/04 14:41:40 INFO DAGScheduler: Missing parents: List()
26/01/04 14:41:40 INFO DAGScheduler: Missing parents found for ResultStage 7: List()
26/01/04 14:41:40 INFO DAGScheduler: Submitting ResultStage 7 (MapPartitionsRDD[24] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:41:40 INFO MemoryStore: Block broadcast_9 stored as values in memory (estimated size 50.6 KiB, free
   434.0 MiB)
26/01/04 14:41:40 INFO MemoryStore: Block broadcast_9_piece0 stored as bytes in memory (estimated size 22.4
  KiB, free 434.0 MiB)
26/01/04 14:41:40 INFO SparkContext: Created broadcast 9 from broadcast at DAGScheduler.scala:1701
26/01/04 14:41:40 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 7 (MapPartitionsRDD[24] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:41:40 INFO TaskSchedulerImpl: Adding task set 7.0 with 1 tasks resource profile 0
26/01/04 14:41:40 INFO TaskSetManager: Starting task 0.0 in stage 7.0 (TID 5) (172.18.32.203,executor 0,
  partition 0, NODE_LOCAL, 9635 bytes)
26/01/04 14:41:41 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 1 to
  172.18.32.203:57392
26/01/04 14:41:41 INFO TaskSetManager: Finished task 0.0 in stage 7.0 (TID 5) in 914 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:41:41 INFO TaskSchedulerImpl: Removed TaskSet 7.0 whose tasks have all completed, from pool
26/01/04 14:41:41 INFO DAGScheduler: ResultStage 7 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 954 ms
26/01/04 14:41:41 INFO DAGScheduler: Job 5 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:41:41 INFO TaskSchedulerImpl: Canceling stage 7
26/01/04 14:41:41 INFO TaskSchedulerImpl: Killing all running tasks in stage 7: Stage finished
26/01/04 14:41:41 INFO DAGScheduler: Job 5 finished: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768, took 1000.076676 ms
26/01/04 14:41:41 INFO CodeGenerator: Code generated in 58.694271 ms
+------+--------------------+---------------+
|Pclass|Survival_Rate_Percent|Total_Passengers|
+------+--------------------+---------------+
|     1|    62.96296296296296|            216|
|     2|    47.28260869565217|            184|
|     3|   24.236252545824847|            491|
+------+--------------------+---------------+


--- Average Age: Survivors vs Non-Survivors ---
26/01/04 14:41:41 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:41:41 INFO FileSourceStrategy: Post-Scan Filters: Set()
26/01/04 14:41:42 INFO CodeGenerator: Code generated in 184.413623 ms
```

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26/01/04 14:41:42 INFO MemoryStore: Block broadcast_10 stored as values in memory (estimated size 215.5 KiB,
  free 433.8 MiB)
26/01/04 14:41:42 INFO MemoryStore: Block broadcast_10_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.8 MiB)
26/01/04 14:41:42 INFO SparkContext: Created broadcast 10 from $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:41:42 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:41:42 INFO DAGScheduler: Registering RDD 28 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 2
26/01/04 14:41:42 INFO DAGScheduler: Got map stage job 6 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:41:42 INFO DAGScheduler: Final stage: ShuffleMapStage 8 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:41:42 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:41:42 INFO DAGScheduler: Missing parents: List()
26/01/04 14:41:42 INFO DAGScheduler: Missing parents found for ShuffleMapStage 8: List()
26/01/04 14:41:42 INFO DAGScheduler: Submitting ShuffleMapStage 8 (MapPartitionsRDD[28] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:41:42 INFO MemoryStore: Block broadcast_11 stored as values in memory (estimated size 41.5 KiB,
  free 433.7 MiB)
26/01/04 14:41:42 INFO MemoryStore: Block broadcast_11_piece0 stored as bytes in memory (estimated size 18.8
  KiB, free 433.7 MiB)
26/01/04 14:41:42 INFO SparkContext: Created broadcast 11 from broadcast at DAGScheduler.scala:1701
26/01/04 14:41:42 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 8 (MapPartitionsRDD[28] at
   $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:41:42 INFO TaskSchedulerImpl: Adding task set 8.0 with 1 tasks resource profile 0
26/01/04 14:41:42 INFO TaskSetManager: Starting task 0.0 in stage 8.0 (TID 6) (172.18.32.200,executor 1,
  partition 0, ANY, 10239 bytes)
26/01/04 14:41:51 INFO TaskSetManager: Finished task 0.0 in stage 8.0 (TID 6) in 9008 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:41:51 INFO TaskSchedulerImpl: Removed TaskSet 8.0 whose tasks have all completed, from pool
26/01/04 14:41:51 INFO DAGScheduler: ShuffleMapStage 8 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 9048 ms
26/01/04 14:41:51 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:41:51 INFO DAGScheduler: running: HashSet()
26/01/04 14:41:51 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:41:51 INFO DAGScheduler: failed: HashSet()
26/01/04 14:41:51 INFO ShufflePartitionsUtil: For shuffle(2, advisory target size: 67108864, actual target size
   1048576, minimum partition size: 1048576
26/01/04 14:41:51 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twolevel.enabled is set to true, but
  current version of codegened fast hashmap does not support this aggregate.
26/01/04 14:41:51 INFO CodeGenerator: Code generated in 93.667647 ms
26/01/04 14:41:51 INFO SparkContext: Starting job: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:41:51 INFO DAGScheduler: Got job 7 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:41:51 INFO DAGScheduler: Final stage: ResultStage 10 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:41:51 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 9)
26/01/04 14:41:51 INFO DAGScheduler: Missing parents: List()
26/01/04 14:41:51 INFO DAGScheduler: Missing parents found for ResultStage 10: List()
26/01/04 14:41:51 INFO DAGScheduler: Submitting ResultStage 10 (MapPartitionsRDD[31] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:41:51 INFO MemoryStore: Block broadcast_12 stored as values in memory (estimated size 48.7 KiB,
  free 433.7 MiB)
26/01/04 14:41:51 INFO MemoryStore: Block broadcast_12_piece0 stored as bytes in memory (estimated size 21.3
  KiB, free 433.6 MiB)
26/01/04 14:41:51 INFO SparkContext: Created broadcast 12 from broadcast at DAGScheduler.scala:1701
26/01/04 14:41:51 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 10 (MapPartitionsRDD[31] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:41:51 INFO TaskSchedulerImpl: Adding task set 10.0 with 1 tasks resource profile 0
26/01/04 14:41:51 INFO TaskSetManager: Starting task 0.0 in stage 10.0 (TID 7) (172.18.32.200,executor 1,
  partition 0, NODE_LOCAL, 9635 bytes)
26/01/04 14:41:52 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 2 to
  172.18.32.200:56416
26/01/04 14:41:52 INFO TaskSetManager: Finished task 0.0 in stage 10.0 (TID 7) in 1373 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:41:52 INFO TaskSchedulerImpl: Removed TaskSet 10.0 whose tasks have all completed, from pool
26/01/04 14:41:52 INFO DAGScheduler: ResultStage 10 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 1423 ms
26/01/04 14:41:52 INFO DAGScheduler: Job 7 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:41:52 INFO TaskSchedulerImpl: Canceling stage 10
26/01/04 14:41:52 INFO TaskSchedulerImpl: Killing all running tasks in stage 10: Stage finished
26/01/04 14:41:52 INFO DAGScheduler: Job 7 finished: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768, took 1462.213905 ms
+--------+-----------------+------------------+
|Survived|       Average_Age|Count_With_Age_Data|
```

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+--------+------------------+-------------------+
|       1|28.343689655172415|                290|
|       0| 30.62617924528302|                424|
+--------+------------------+-------------------+


--- Starting Machine Learning Pipeline ---
26/01/04 14:41:56 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:41:56 INFO FileSourceStrategy: Post-Scan Filters: Set(atleastnnonnulls(6, Pclass#19, Sex#21,
  Age#22, Fare#26, Embarked#28, Survived#18))
26/01/04 14:41:56 INFO CodeGenerator: Code generated in 147.885941 ms
26/01/04 14:41:56 INFO MemoryStore: Block broadcast_13 stored as values in memory (estimated size 215.5 KiB,
  free 433.9 MiB)
26/01/04 14:41:56 INFO MemoryStore: Block broadcast_13_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 434.2 MiB)
26/01/04 14:41:56 INFO SparkContext: Created broadcast 13 from $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:41:56 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:41:56 INFO DAGScheduler: Registering RDD 35 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 3
26/01/04 14:41:56 INFO DAGScheduler: Got map stage job 8 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:41:56 INFO DAGScheduler: Final stage: ShuffleMapStage 11 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:41:56 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:41:56 INFO DAGScheduler: Missing parents: List()
26/01/04 14:41:56 INFO DAGScheduler: Missing parents found for ShuffleMapStage 11: List()
26/01/04 14:41:56 INFO DAGScheduler: Submitting ShuffleMapStage 11 (MapPartitionsRDD[35] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:41:56 INFO MemoryStore: Block broadcast_14 stored as values in memory (estimated size 43.9 KiB,
  free 434.1 MiB)
26/01/04 14:41:56 INFO MemoryStore: Block broadcast_14_piece0 stored as bytes in memory (estimated size 19.4
  KiB, free 434.1 MiB)
26/01/04 14:41:56 INFO SparkContext: Created broadcast 14 from broadcast at DAGScheduler.scala:1701
26/01/04 14:41:56 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 11 (MapPartitionsRDD[35]
  at $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:41:56 INFO TaskSchedulerImpl: Adding task set 11.0 with 1 tasks resource profile 0
26/01/04 14:41:56 INFO TaskSetManager: Starting task 0.0 in stage 11.0 (TID 8) (172.18.32.200,executor 1,
  partition 0, ANY, 10239 bytes)
26/01/04 14:41:58 INFO TaskSetManager: Finished task 0.0 in stage 11.0 (TID 8) in 1545 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:41:58 INFO TaskSchedulerImpl: Removed TaskSet 11.0 whose tasks have all completed, from pool
26/01/04 14:41:58 INFO DAGScheduler: ShuffleMapStage 11 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 1595 ms
26/01/04 14:41:58 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:41:58 INFO DAGScheduler: running: HashSet()
26/01/04 14:41:58 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:41:58 INFO DAGScheduler: failed: HashSet()
26/01/04 14:41:58 INFO ShufflePartitionsUtil: For shuffle(3, advisory target size: 67108864, actual target size
   1048576, minimum partition size: 1048576
26/01/04 14:41:58 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twolevel.enabled is set to true, but
  current version of codegened fast hashmap does not support this aggregate.
26/01/04 14:41:58 INFO CodeGenerator: Code generated in 66.714063 ms
26/01/04 14:41:58 INFO DAGScheduler: Registering RDD 39 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 4
26/01/04 14:41:58 INFO DAGScheduler: Got map stage job 9 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:41:58 INFO DAGScheduler: Final stage: ShuffleMapStage 13 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:41:58 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 12)
26/01/04 14:41:58 INFO DAGScheduler: Missing parents: List()
26/01/04 14:41:58 INFO DAGScheduler: Missing parents found for ShuffleMapStage 13: List()
26/01/04 14:41:58 INFO DAGScheduler: Submitting ShuffleMapStage 13 (MapPartitionsRDD[39] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:41:58 INFO MemoryStore: Block broadcast_15 stored as values in memory (estimated size 49.1 KiB,
  free 434.0 MiB)
26/01/04 14:41:58 INFO MemoryStore: Block broadcast_15_piece0 stored as bytes in memory (estimated size 21.5
  KiB, free 434.0 MiB)
26/01/04 14:41:58 INFO SparkContext: Created broadcast 15 from broadcast at DAGScheduler.scala:1701
26/01/04 14:41:58 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 13 (MapPartitionsRDD[39]
  at $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:41:58 INFO TaskSchedulerImpl: Adding task set 13.0 with 1 tasks resource profile 0
26/01/04 14:41:58 INFO TaskSetManager: Starting task 0.0 in stage 13.0 (TID 9) (172.18.32.200,executor 1,
  partition 0, NODE_LOCAL, 9624 bytes)
26/01/04 14:41:58 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 3 to
  172.18.32.200:56416
26/01/04 14:41:59 INFO TaskSetManager: Finished task 0.0 in stage 13.0 (TID 9) in 702 ms on 172.18.32.200
  (executor 1) (1/1)
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26/01/04 14:41:59 INFO TaskSchedulerImpl: Removed TaskSet 13.0 whose tasks have all completed, from pool
26/01/04 14:41:59 INFO DAGScheduler: ShuffleMapStage 13 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 748 ms
26/01/04 14:41:59 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:41:59 INFO DAGScheduler: running: HashSet()
26/01/04 14:41:59 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:41:59 INFO DAGScheduler: failed: HashSet()
26/01/04 14:41:59 INFO ShufflePartitionsUtil: For shuffle(4, advisory target size: 67108864, actual target size
   1048576, minimum partition size: 1048576
26/01/04 14:41:59 INFO SparkContext: Starting job: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:41:59 INFO DAGScheduler: Got job 10 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:41:59 INFO DAGScheduler: Final stage: ResultStage 16 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:41:59 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 15)
26/01/04 14:41:59 INFO DAGScheduler: Missing parents: List()
26/01/04 14:41:59 INFO DAGScheduler: Missing parents found for ResultStage 16: List()
26/01/04 14:41:59 INFO DAGScheduler: Submitting ResultStage 16 (MapPartitionsRDD[42] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:41:59 INFO MemoryStore: Block broadcast_16 stored as values in memory (estimated size 48.0 KiB,
  free 434.0 MiB)
26/01/04 14:41:59 INFO MemoryStore: Block broadcast_16_piece0 stored as bytes in memory (estimated size 20.5
  KiB, free 434.0 MiB)
26/01/04 14:41:59 INFO SparkContext: Created broadcast 16 from broadcast at DAGScheduler.scala:1701
26/01/04 14:41:59 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 16 (MapPartitionsRDD[42] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:41:59 INFO TaskSchedulerImpl: Adding task set 16.0 with 1 tasks resource profile 0
26/01/04 14:41:59 INFO TaskSetManager: Starting task 0.0 in stage 16.0 (TID 10) (172.18.32.200,executor 1,
  partition 0, NODE_LOCAL, 9635 bytes)
26/01/04 14:41:59 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 4 to
  172.18.32.200:56416
26/01/04 14:42:00 INFO TaskSetManager: Finished task 0.0 in stage 16.0 (TID 10) in 798 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:42:00 INFO TaskSchedulerImpl: Removed TaskSet 16.0 whose tasks have all completed, from pool
26/01/04 14:42:00 INFO DAGScheduler: ResultStage 16 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 843 ms
26/01/04 14:42:00 INFO DAGScheduler: Job 10 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:00 INFO TaskSchedulerImpl: Canceling stage 16
26/01/04 14:42:00 INFO TaskSchedulerImpl: Killing all running tasks in stage 16: Stage finished
26/01/04 14:42:00 INFO DAGScheduler: Job 10 finished: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768, took 893.888503 ms
26/01/04 14:42:00 INFO CodeGenerator: Code generated in 87.968325 ms
26/01/04 14:42:01 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:42:01 INFO FileSourceStrategy: Post-Scan Filters: Set(atleastnnonnulls(6, Pclass#19, Sex#21,
  Age#22, Fare#26, Embarked#28, Survived#18))
26/01/04 14:42:01 INFO CodeGenerator: Code generated in 150.50315 ms
26/01/04 14:42:01 INFO MemoryStore: Block broadcast_17 stored as values in memory (estimated size 215.5 KiB,
  free 433.7 MiB)
26/01/04 14:42:02 INFO MemoryStore: Block broadcast_17_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.7 MiB)
26/01/04 14:42:02 INFO SparkContext: Created broadcast 17 from $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:42:02 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:42:02 INFO DAGScheduler: Registering RDD 46 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 5
26/01/04 14:42:02 INFO DAGScheduler: Got map stage job 11 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:42:02 INFO DAGScheduler: Final stage: ShuffleMapStage 17 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:42:02 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:02 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:02 INFO DAGScheduler: Missing parents found for ShuffleMapStage 17: List()
26/01/04 14:42:02 INFO DAGScheduler: Submitting ShuffleMapStage 17 (MapPartitionsRDD[46] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:42:02 INFO MemoryStore: Block broadcast_18 stored as values in memory (estimated size 44.0 KiB,
  free 433.7 MiB)
26/01/04 14:42:02 INFO MemoryStore: Block broadcast_18_piece0 stored as bytes in memory (estimated size 19.4
  KiB, free 433.6 MiB)
26/01/04 14:42:02 INFO SparkContext: Created broadcast 18 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:02 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 17 (MapPartitionsRDD[46]
  at $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:42:02 INFO TaskSchedulerImpl: Adding task set 17.0 with 1 tasks resource profile 0
26/01/04 14:42:02 INFO TaskSetManager: Starting task 0.0 in stage 17.0 (TID 11) (172.18.32.200,executor 1,
  partition 0, ANY, 10239 bytes)
26/01/04 14:42:02 INFO TaskSetManager: Finished task 0.0 in stage 17.0 (TID 11) in 662 ms on 172.18.32.200
  (executor 1) (1/1)
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26/01/04 14:42:02 INFO TaskSchedulerImpl: Removed TaskSet 17.0 whose tasks have all completed, from pool
26/01/04 14:42:02 INFO DAGScheduler: ShuffleMapStage 17 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 692 ms
26/01/04 14:42:02 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:42:02 INFO DAGScheduler: running: HashSet()
26/01/04 14:42:02 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:42:02 INFO DAGScheduler: failed: HashSet()
26/01/04 14:42:02 INFO ShufflePartitionsUtil: For shuffle(5, advisory target size: 67108864, actual target size
   1048576, minimum partition size: 1048576
26/01/04 14:42:02 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twolevel.enabled is set to true, but
  current version of codegened fast hashmap does not support this aggregate.
26/01/04 14:42:02 INFO DAGScheduler: Registering RDD 50 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 6
26/01/04 14:42:02 INFO DAGScheduler: Got map stage job 12 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:42:02 INFO DAGScheduler: Final stage: ShuffleMapStage 19 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:42:02 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 18)
26/01/04 14:42:02 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:02 INFO DAGScheduler: Missing parents found for ShuffleMapStage 19: List()
26/01/04 14:42:02 INFO DAGScheduler: Submitting ShuffleMapStage 19 (MapPartitionsRDD[50] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:42:02 INFO MemoryStore: Block broadcast_19 stored as values in memory (estimated size 49.1 KiB,
  free 433.6 MiB)
26/01/04 14:42:02 INFO MemoryStore: Block broadcast_19_piece0 stored as bytes in memory (estimated size 21.5
  KiB, free 433.6 MiB)
26/01/04 14:42:02 INFO SparkContext: Created broadcast 19 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:02 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 19 (MapPartitionsRDD[50]
  at $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:42:02 INFO TaskSchedulerImpl: Adding task set 19.0 with 1 tasks resource profile 0
26/01/04 14:42:02 INFO TaskSetManager: Starting task 0.0 in stage 19.0 (TID 12) (172.18.32.200,executor 1,
  partition 0, NODE_LOCAL, 9624 bytes)
26/01/04 14:42:03 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 5 to
  172.18.32.200:56416
26/01/04 14:42:03 INFO TaskSetManager: Finished task 0.0 in stage 19.0 (TID 12) in 325 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:42:03 INFO TaskSchedulerImpl: Removed TaskSet 19.0 whose tasks have all completed, from pool
26/01/04 14:42:03 INFO DAGScheduler: ShuffleMapStage 19 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 366 ms
26/01/04 14:42:03 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:42:03 INFO DAGScheduler: running: HashSet()
26/01/04 14:42:03 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:42:03 INFO DAGScheduler: failed: HashSet()
26/01/04 14:42:03 INFO ShufflePartitionsUtil: For shuffle(6, advisory target size: 67108864, actual target size
   1048576, minimum partition size: 1048576
26/01/04 14:42:03 INFO SparkContext: Starting job: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:42:03 INFO DAGScheduler: Got job 13 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:42:03 INFO DAGScheduler: Final stage: ResultStage 22 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:42:03 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 21)
26/01/04 14:42:03 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:03 INFO DAGScheduler: Missing parents found for ResultStage 22: List()
26/01/04 14:42:03 INFO DAGScheduler: Submitting ResultStage 22 (MapPartitionsRDD[53] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:42:03 INFO MemoryStore: Block broadcast_20 stored as values in memory (estimated size 48.0 KiB,
  free 433.7 MiB)
26/01/04 14:42:03 INFO MemoryStore: Block broadcast_20_piece0 stored as bytes in memory (estimated size 20.4
  KiB, free 433.7 MiB)
26/01/04 14:42:03 INFO SparkContext: Created broadcast 20 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:03 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 22 (MapPartitionsRDD[53] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:42:03 INFO TaskSchedulerImpl: Adding task set 22.0 with 1 tasks resource profile 0
26/01/04 14:42:03 INFO TaskSetManager: Starting task 0.0 in stage 22.0 (TID 13) (172.18.32.200,executor 1,
  partition 0, NODE_LOCAL, 9635 bytes)
26/01/04 14:42:03 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 6 to
  172.18.32.200:56416
26/01/04 14:42:03 INFO TaskSetManager: Finished task 0.0 in stage 22.0 (TID 13) in 276 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:42:03 INFO TaskSchedulerImpl: Removed TaskSet 22.0 whose tasks have all completed, from pool
26/01/04 14:42:03 INFO DAGScheduler: ResultStage 22 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 325 ms
26/01/04 14:42:03 INFO DAGScheduler: Job 13 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:03 INFO TaskSchedulerImpl: Canceling stage 22
26/01/04 14:42:03 INFO TaskSchedulerImpl: Killing all running tasks in stage 22: Stage finished
26/01/04 14:42:03 INFO DAGScheduler: Job 13 finished: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768, took 354.231136 ms
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26/01/04 14:42:04 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:42:04 INFO FileSourceStrategy: Post-Scan Filters: Set(atleastnnonnulls(6, Pclass#19, Sex#21,
  Age#22, Fare#26, Embarked#28, Survived#18))
26/01/04 14:42:05 INFO CodeGenerator: Code generated in 245.517512 ms
26/01/04 14:42:05 INFO MemoryStore: Block broadcast_21 stored as values in memory (estimated size 215.5 KiB,
  free 433.9 MiB)
26/01/04 14:42:05 INFO MemoryStore: Block broadcast_21_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.8 MiB)
26/01/04 14:42:05 INFO SparkContext: Created broadcast 21 from $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:42:05 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:42:05 INFO DAGScheduler: Registering RDD 57 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 7
26/01/04 14:42:05 INFO DAGScheduler: Got map stage job 14 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:42:05 INFO DAGScheduler: Final stage: ShuffleMapStage 23 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:42:05 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:05 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:05 INFO DAGScheduler: Missing parents found for ShuffleMapStage 23: List()
26/01/04 14:42:05 INFO DAGScheduler: Submitting ShuffleMapStage 23 (MapPartitionsRDD[57] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:42:05 INFO MemoryStore: Block broadcast_22 stored as values in memory (estimated size 65.2 KiB,
  free 433.8 MiB)
26/01/04 14:42:05 INFO MemoryStore: Block broadcast_22_piece0 stored as bytes in memory (estimated size 27.2
  KiB, free 433.7 MiB)
26/01/04 14:42:05 INFO SparkContext: Created broadcast 22 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:05 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 23 (MapPartitionsRDD[57]
  at $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:42:05 INFO TaskSchedulerImpl: Adding task set 23.0 with 1 tasks resource profile 0
26/01/04 14:42:05 INFO TaskSetManager: Starting task 0.0 in stage 23.0 (TID 14) (172.18.32.200,executor 1,
  partition 0, ANY, 10239 bytes)
26/01/04 14:42:11 INFO TaskSetManager: Finished task 0.0 in stage 23.0 (TID 14) in 5952 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:42:11 INFO TaskSchedulerImpl: Removed TaskSet 23.0 whose tasks have all completed, from pool
26/01/04 14:42:11 INFO DAGScheduler: ShuffleMapStage 23 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 5986 ms
26/01/04 14:42:11 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:42:11 INFO DAGScheduler: running: HashSet()
26/01/04 14:42:11 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:42:11 INFO DAGScheduler: failed: HashSet()
26/01/04 14:42:11 INFO CodeGenerator: Code generated in 55.938693 ms
26/01/04 14:42:11 INFO SparkContext: Starting job: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:42:11 INFO DAGScheduler: Got job 15 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:42:11 INFO DAGScheduler: Final stage: ResultStage 25 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:42:11 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 24)
26/01/04 14:42:11 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:11 INFO DAGScheduler: Missing parents found for ResultStage 25: List()
26/01/04 14:42:11 INFO DAGScheduler: Submitting ResultStage 25 (MapPartitionsRDD[60] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:42:11 INFO MemoryStore: Block broadcast_23 stored as values in memory (estimated size 13.8 KiB,
  free 433.7 MiB)
26/01/04 14:42:11 INFO MemoryStore: Block broadcast_23_piece0 stored as bytes in memory (estimated size 6.6
  KiB, free 433.7 MiB)
26/01/04 14:42:11 INFO SparkContext: Created broadcast 23 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:11 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 25 (MapPartitionsRDD[60] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:42:11 INFO TaskSchedulerImpl: Adding task set 25.0 with 1 tasks resource profile 0
26/01/04 14:42:11 INFO TaskSetManager: Starting task 0.0 in stage 25.0 (TID 15) (172.18.32.200,executor 1,
  partition 0, NODE_LOCAL, 9635 bytes)
26/01/04 14:42:11 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 7 to
  172.18.32.200:56416
26/01/04 14:42:11 INFO TaskSetManager: Finished task 0.0 in stage 25.0 (TID 15) in 222 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:42:11 INFO TaskSchedulerImpl: Removed TaskSet 25.0 whose tasks have all completed, from pool
26/01/04 14:42:11 INFO DAGScheduler: ResultStage 25 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 282 ms
26/01/04 14:42:11 INFO DAGScheduler: Job 15 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:11 INFO TaskSchedulerImpl: Canceling stage 25
26/01/04 14:42:11 INFO TaskSchedulerImpl: Killing all running tasks in stage 25: Stage finished
26/01/04 14:42:11 INFO DAGScheduler: Job 15 finished: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768, took 336.278729 ms
Training Dataset Count: 598
26/01/04 14:42:12 INFO FileSourceStrategy: Pushed Filters:
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26/01/04 14:42:12 INFO FileSourceStrategy: Post-Scan Filters: Set(atleastnnonnulls(6, Pclass#19, Sex#21,
  Age#22, Fare#26, Embarked#28, Survived#18))
26/01/04 14:42:12 INFO CodeGenerator: Code generated in 177.021828 ms
26/01/04 14:42:12 INFO MemoryStore: Block broadcast_24 stored as values in memory (estimated size 215.5 KiB,
  free 433.5 MiB)
26/01/04 14:42:12 INFO MemoryStore: Block broadcast_24_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.5 MiB)
26/01/04 14:42:12 INFO SparkContext: Created broadcast 24 from $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:42:12 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:42:12 INFO DAGScheduler: Registering RDD 64 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) as input to shuffle 8
26/01/04 14:42:12 INFO DAGScheduler: Got map stage job 16 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:42:12 INFO DAGScheduler: Final stage: ShuffleMapStage 26 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:42:12 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:12 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:12 INFO DAGScheduler: Missing parents found for ShuffleMapStage 26: List()
26/01/04 14:42:12 INFO DAGScheduler: Submitting ShuffleMapStage 26 (MapPartitionsRDD[64] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:42:12 INFO MemoryStore: Block broadcast_25 stored as values in memory (estimated size 65.2 KiB,
  free 433.4 MiB)
26/01/04 14:42:12 INFO MemoryStore: Block broadcast_25_piece0 stored as bytes in memory (estimated size 27.2
  KiB, free 433.4 MiB)
26/01/04 14:42:12 INFO SparkContext: Created broadcast 25 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:12 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 26 (MapPartitionsRDD[64]
  at $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:42:12 INFO TaskSchedulerImpl: Adding task set 26.0 with 1 tasks resource profile 0
26/01/04 14:42:12 INFO TaskSetManager: Starting task 0.0 in stage 26.0 (TID 16) (172.18.32.200,executor 1,
  partition 0, ANY, 10239 bytes)
26/01/04 14:42:14 INFO TaskSetManager: Finished task 0.0 in stage 26.0 (TID 16) in 1452 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:42:14 INFO TaskSchedulerImpl: Removed TaskSet 26.0 whose tasks have all completed, from pool
26/01/04 14:42:14 INFO DAGScheduler: ShuffleMapStage 26 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 1493 ms
26/01/04 14:42:14 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:42:14 INFO DAGScheduler: running: HashSet()
26/01/04 14:42:14 INFO DAGScheduler: waiting: HashSet()
26/01/04 14:42:14 INFO DAGScheduler: failed: HashSet()
26/01/04 14:42:14 INFO SparkContext: Starting job: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768
26/01/04 14:42:14 INFO DAGScheduler: Got job 17 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) with 1 output partitions
26/01/04 14:42:14 INFO DAGScheduler: Final stage: ResultStage 28 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768)
26/01/04 14:42:14 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 27)
26/01/04 14:42:14 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:14 INFO DAGScheduler: Missing parents found for ResultStage 28: List()
26/01/04 14:42:14 INFO DAGScheduler: Submitting ResultStage 28 (MapPartitionsRDD[67] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768), which has no missing parents
26/01/04 14:42:14 INFO MemoryStore: Block broadcast_26 stored as values in memory (estimated size 13.8 KiB,
  free 433.5 MiB)
26/01/04 14:42:14 INFO MemoryStore: Block broadcast_26_piece0 stored as bytes in memory (estimated size 6.6
  KiB, free 433.5 MiB)
26/01/04 14:42:14 INFO SparkContext: Created broadcast 26 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:14 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 28 (MapPartitionsRDD[67] at
  $anonfun$withThreadLocalCaptured$2 at CompletableFuture.java:1768) (first 15 tasks are for partitions
  Vector(0))
26/01/04 14:42:14 INFO TaskSchedulerImpl: Adding task set 28.0 with 1 tasks resource profile 0
26/01/04 14:42:14 INFO TaskSetManager: Starting task 0.0 in stage 28.0 (TID 17) (172.18.32.200,executor 1,
  partition 0, NODE_LOCAL, 9635 bytes)
26/01/04 14:42:14 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 8 to
  172.18.32.200:56416
26/01/04 14:42:14 INFO TaskSetManager: Finished task 0.0 in stage 28.0 (TID 17) in 154 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:42:14 INFO TaskSchedulerImpl: Removed TaskSet 28.0 whose tasks have all completed, from pool
26/01/04 14:42:14 INFO DAGScheduler: ResultStage 28 ($anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768) finished in 219 ms
26/01/04 14:42:14 INFO DAGScheduler: Job 17 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:14 INFO TaskSchedulerImpl: Canceling stage 28
26/01/04 14:42:14 INFO TaskSchedulerImpl: Killing all running tasks in stage 28: Stage finished
26/01/04 14:42:14 INFO DAGScheduler: Job 17 finished: $anonfun$withThreadLocalCaptured$2 at
  CompletableFuture.java:1768, took 254.80239 ms
Test Dataset Count: 114
26/01/04 14:42:15 INFO Instrumentation: [987f2c47] Stage class: LogisticRegression
26/01/04 14:42:15 INFO Instrumentation: [987f2c47] Stage uid: LogisticRegression_0d7f83a6eace
26/01/04 14:42:15 INFO FileSourceStrategy: Pushed Filters:
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26/01/04 14:42:15 INFO FileSourceStrategy: Post-Scan Filters: Set(atleastnnonnulls(6, Pclass#19, Sex#21,
  Age#22, Fare#26, Embarked#28, Survived#18))
26/01/04 14:42:15 INFO CodeGenerator: Code generated in 213.563307 ms
26/01/04 14:42:15 INFO MemoryStore: Block broadcast_27 stored as values in memory (estimated size 215.5 KiB,
  free 433.9 MiB)
26/01/04 14:42:15 INFO MemoryStore: Block broadcast_27_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.9 MiB)
26/01/04 14:42:15 INFO SparkContext: Created broadcast 27 from rdd at Instrumentation.scala:64
26/01/04 14:42:15 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:42:15 INFO Instrumentation: [987f2c47] training: numPartitions=1 storageLevel=StorageLevel(1
  replicas)
26/01/04 14:42:15 INFO Instrumentation: [987f2c47] {"labelCol":"Survived","featuresCol":"features"}
26/01/04 14:42:16 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:42:16 INFO FileSourceStrategy: Post-Scan Filters: Set(atleastnnonnulls(6, Pclass#19, Sex#21,
  Age#22, Fare#26, Embarked#28, Survived#18))
26/01/04 14:42:16 INFO CodeGenerator: Code generated in 251.396371 ms
26/01/04 14:42:16 INFO MemoryStore: Block broadcast_28 stored as values in memory (estimated size 215.5 KiB,
  free 433.7 MiB)
26/01/04 14:42:17 INFO MemoryStore: Block broadcast_28_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.6 MiB)
26/01/04 14:42:17 INFO SparkContext: Created broadcast 28 from rdd at LogisticRegression.scala:522
26/01/04 14:42:17 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:42:17 INFO SparkContext: Starting job: treeAggregate at Summarizer.scala:238
26/01/04 14:42:17 INFO DAGScheduler: Got job 18 (treeAggregate at Summarizer.scala:238) with 1 output
  partitions
26/01/04 14:42:17 INFO DAGScheduler: Final stage: ResultStage 29 (treeAggregate at Summarizer.scala:238)
26/01/04 14:42:17 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:17 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:17 INFO DAGScheduler: Missing parents found for ResultStage 29: List()
26/01/04 14:42:17 INFO DAGScheduler: Submitting ResultStage 29 (MapPartitionsRDD[82] at treeAggregate at
  Summarizer.scala:238), which has no missing parents
26/01/04 14:42:17 INFO MemoryStore: Block broadcast_29 stored as values in memory (estimated size 89.9 KiB,
  free 433.5 MiB)
26/01/04 14:42:17 INFO MemoryStore: Block broadcast_29_piece0 stored as bytes in memory (estimated size 35.0
  KiB, free 433.5 MiB)
26/01/04 14:42:17 INFO SparkContext: Created broadcast 29 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:17 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 29 (MapPartitionsRDD[82] at
  treeAggregate at Summarizer.scala:238) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:17 INFO TaskSchedulerImpl: Adding task set 29.0 with 1 tasks resource profile 0
26/01/04 14:42:17 INFO TaskSetManager: Starting task 0.0 in stage 29.0 (TID 18) (172.18.32.203,executor 0,
  partition 0, ANY, 10250 bytes)
26/01/04 14:42:23 INFO TaskSetManager: Finished task 0.0 in stage 29.0 (TID 18) in 6107 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:23 INFO TaskSchedulerImpl: Removed TaskSet 29.0 whose tasks have all completed, from pool
26/01/04 14:42:23 INFO DAGScheduler: ResultStage 29 (treeAggregate at Summarizer.scala:238) finished in 6265 ms
26/01/04 14:42:23 INFO DAGScheduler: Job 18 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:23 INFO TaskSchedulerImpl: Canceling stage 29
26/01/04 14:42:23 INFO TaskSchedulerImpl: Killing all running tasks in stage 29: Stage finished
26/01/04 14:42:23 INFO DAGScheduler: Job 18 finished: treeAggregate at Summarizer.scala:238, took 6281.422368
  ms
26/01/04 14:42:23 INFO Instrumentation: [987f2c47] {"numClasses":2}
26/01/04 14:42:23 INFO Instrumentation: [987f2c47] {"numFeatures":5}
26/01/04 14:42:23 INFO Instrumentation: [987f2c47] {"numExamples":598}
26/01/04 14:42:23 INFO Instrumentation: [987f2c47] {"lowestLabelWeight":"237.0"}
26/01/04 14:42:23 INFO Instrumentation: [987f2c47] {"highestLabelWeight":"361.0"}
26/01/04 14:42:23 INFO Instrumentation: [987f2c47] {"sumOfWeights":598.0}
26/01/04 14:42:23 INFO Instrumentation: [987f2c47] {"actualBlockSizeInMB":"1.0"}
26/01/04 14:42:29 INFO MemoryStore: Block broadcast_30 stored as values in memory (estimated size 80.0 B, free
  433.9 MiB)
26/01/04 14:42:29 INFO MemoryStore: Block broadcast_30_piece0 stored as bytes in memory (estimated size 109.0
  B, free 433.9 MiB)
26/01/04 14:42:29 INFO SparkContext: Created broadcast 30 from broadcast at LogisticRegression.scala:954
26/01/04 14:42:29 INFO MemoryStore: Block broadcast_31 stored as values in memory (estimated size 80.0 B, free
  433.9 MiB)
26/01/04 14:42:29 INFO MemoryStore: Block broadcast_31_piece0 stored as bytes in memory (estimated size 109.0
  B, free 433.9 MiB)
26/01/04 14:42:29 INFO SparkContext: Created broadcast 31 from broadcast at LogisticRegression.scala:955
26/01/04 14:42:30 INFO MemoryStore: Block broadcast_32 stored as values in memory (estimated size 104.0 B, free
   433.9 MiB)
26/01/04 14:42:30 INFO MemoryStore: Block broadcast_32_piece0 stored as bytes in memory (estimated size 154.0
  B, free 433.9 MiB)
26/01/04 14:42:30 INFO SparkContext: Created broadcast 32 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:30 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:30 INFO DAGScheduler: Got job 19 (treeAggregate at RDDLossFunction.scala:61) with 1 output
  partitions
26/01/04 14:42:30 INFO DAGScheduler: Final stage: ResultStage 30 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:30 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:30 INFO DAGScheduler: Missing parents: List()
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26/01/04 14:42:30 INFO DAGScheduler: Missing parents found for ResultStage 30: List()
26/01/04 14:42:30 INFO DAGScheduler: Submitting ResultStage 30 (MapPartitionsRDD[85] at treeAggregate at
  RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:30 INFO MemoryStore: Block broadcast_33 stored as values in memory (estimated size 90.5 KiB,
  free 433.8 MiB)
26/01/04 14:42:30 INFO MemoryStore: Block broadcast_33_piece0 stored as bytes in memory (estimated size 34.9
  KiB, free 433.8 MiB)
26/01/04 14:42:30 INFO SparkContext: Created broadcast 33 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:30 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 30 (MapPartitionsRDD[85] at
  treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:30 INFO TaskSchedulerImpl: Adding task set 30.0 with 1 tasks resource profile 0
26/01/04 14:42:30 INFO TaskSetManager: Starting task 0.0 in stage 30.0 (TID 19) (172.18.32.203,executor 0,
  partition 0, ANY, 10250 bytes)
26/01/04 14:42:32 INFO TaskSetManager: Finished task 0.0 in stage 30.0 (TID 19) in 1826 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:32 INFO TaskSchedulerImpl: Removed TaskSet 30.0 whose tasks have all completed, from pool
26/01/04 14:42:32 INFO DAGScheduler: ResultStage 30 (treeAggregate at RDDLossFunction.scala:61) finished in
  1858 ms
26/01/04 14:42:32 INFO DAGScheduler: Job 19 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:32 INFO TaskSchedulerImpl: Canceling stage 30
26/01/04 14:42:32 INFO TaskSchedulerImpl: Killing all running tasks in stage 30: Stage finished
26/01/04 14:42:32 INFO DAGScheduler: Job 19 finished: treeAggregate at RDDLossFunction.scala:61, took
  1879.929257 ms
26/01/04 14:42:32 INFO TorrentBroadcast: Destroying Broadcast(32) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:32 INFO MemoryStore: Block broadcast_34 stored as values in memory (estimated size 104.0 B, free
   433.8 MiB)
26/01/04 14:42:32 INFO MemoryStore: Block broadcast_34_piece0 stored as bytes in memory (estimated size 186.0
  B, free 433.8 MiB)
26/01/04 14:42:32 INFO SparkContext: Created broadcast 34 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:32 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:32 INFO DAGScheduler: Got job 20 (treeAggregate at RDDLossFunction.scala:61) with 1 output
  partitions
26/01/04 14:42:32 INFO DAGScheduler: Final stage: ResultStage 31 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:32 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:32 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:32 INFO DAGScheduler: Missing parents found for ResultStage 31: List()
26/01/04 14:42:32 INFO DAGScheduler: Submitting ResultStage 31 (MapPartitionsRDD[86] at treeAggregate at
  RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:32 INFO MemoryStore: Block broadcast_35 stored as values in memory (estimated size 90.5 KiB,
  free 433.7 MiB)
26/01/04 14:42:32 INFO MemoryStore: Block broadcast_35_piece0 stored as bytes in memory (estimated size 34.9
  KiB, free 433.7 MiB)
26/01/04 14:42:32 INFO SparkContext: Created broadcast 35 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:32 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 31 (MapPartitionsRDD[86] at
  treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:32 INFO TaskSchedulerImpl: Adding task set 31.0 with 1 tasks resource profile 0
26/01/04 14:42:32 INFO TaskSetManager: Starting task 0.0 in stage 31.0 (TID 20) (172.18.32.203,executor 0,
  partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:32 INFO TaskSetManager: Finished task 0.0 in stage 31.0 (TID 20) in 232 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:32 INFO TaskSchedulerImpl: Removed TaskSet 31.0 whose tasks have all completed, from pool
26/01/04 14:42:32 INFO DAGScheduler: ResultStage 31 (treeAggregate at RDDLossFunction.scala:61) finished in 259
   ms
26/01/04 14:42:32 INFO DAGScheduler: Job 20 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:32 INFO TaskSchedulerImpl: Canceling stage 31
26/01/04 14:42:32 INFO TaskSchedulerImpl: Killing all running tasks in stage 31: Stage finished
26/01/04 14:42:32 INFO DAGScheduler: Job 20 finished: treeAggregate at RDDLossFunction.scala:61, took
  273.421239 ms
26/01/04 14:42:32 INFO TorrentBroadcast: Destroying Broadcast(34) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:32 INFO LBFGS: Step Size: 2.925
26/01/04 14:42:32 INFO LBFGS: Val and Grad Norm: 0.482006 (rel: 0.282) 0.0918621
26/01/04 14:42:32 INFO MemoryStore: Block broadcast_36 stored as values in memory (estimated size 104.0 B, free
   433.7 MiB)
26/01/04 14:42:32 INFO MemoryStore: Block broadcast_36_piece0 stored as bytes in memory (estimated size 186.0
  B, free 433.7 MiB)
26/01/04 14:42:32 INFO SparkContext: Created broadcast 36 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:32 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:32 INFO DAGScheduler: Got job 21 (treeAggregate at RDDLossFunction.scala:61) with 1 output
  partitions
26/01/04 14:42:32 INFO DAGScheduler: Final stage: ResultStage 32 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:32 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:32 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:32 INFO DAGScheduler: Missing parents found for ResultStage 32: List()
26/01/04 14:42:32 INFO DAGScheduler: Submitting ResultStage 32 (MapPartitionsRDD[87] at treeAggregate at
  RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:32 INFO MemoryStore: Block broadcast_37 stored as values in memory (estimated size 90.5 KiB,
  free 433.6 MiB)
26/01/04 14:42:32 INFO MemoryStore: Block broadcast_37_piece0 stored as bytes in memory (estimated size 34.9
  KiB, free 433.5 MiB)
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26/01/04 14:42:32 INFO SparkContext: Created broadcast 37 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:32 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 32 (MapPartitionsRDD[87] at
  treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:32 INFO TaskSchedulerImpl: Adding task set 32.0 with 1 tasks resource profile 0
26/01/04 14:42:32 INFO TaskSetManager: Starting task 0.0 in stage 32.0 (TID 21) (172.18.32.203,executor 0,
  partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:32 INFO TaskSetManager: Finished task 0.0 in stage 32.0 (TID 21) in 225 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:32 INFO TaskSchedulerImpl: Removed TaskSet 32.0 whose tasks have all completed, from pool
26/01/04 14:42:32 INFO DAGScheduler: ResultStage 32 (treeAggregate at RDDLossFunction.scala:61) finished in 257
   ms
26/01/04 14:42:32 INFO DAGScheduler: Job 21 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:32 INFO TaskSchedulerImpl: Canceling stage 32
26/01/04 14:42:32 INFO TaskSchedulerImpl: Killing all running tasks in stage 32: Stage finished
26/01/04 14:42:32 INFO DAGScheduler: Job 21 finished: treeAggregate at RDDLossFunction.scala:61, took
  271.725822 ms
26/01/04 14:42:32 INFO TorrentBroadcast: Destroying Broadcast(36) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:32 INFO LBFGS: Step Size: 1.000
26/01/04 14:42:32 INFO LBFGS: Val and Grad Norm: 0.458707 (rel: 0.0483) 0.0391740
26/01/04 14:42:32 INFO MemoryStore: Block broadcast_38 stored as values in memory (estimated size 104.0 B, free
   433.5 MiB)
26/01/04 14:42:32 INFO MemoryStore: Block broadcast_38_piece0 stored as bytes in memory (estimated size 186.0
  B, free 433.5 MiB)
26/01/04 14:42:32 INFO SparkContext: Created broadcast 38 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:32 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:32 INFO DAGScheduler: Got job 22 (treeAggregate at RDDLossFunction.scala:61) with 1 output
  partitions
26/01/04 14:42:32 INFO DAGScheduler: Final stage: ResultStage 33 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:32 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:32 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:32 INFO DAGScheduler: Missing parents found for ResultStage 33: List()
26/01/04 14:42:32 INFO DAGScheduler: Submitting ResultStage 33 (MapPartitionsRDD[88] at treeAggregate at
  RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:32 INFO MemoryStore: Block broadcast_39 stored as values in memory (estimated size 90.5 KiB,
  free 433.6 MiB)
26/01/04 14:42:33 INFO MemoryStore: Block broadcast_39_piece0 stored as bytes in memory (estimated size 34.9
  KiB, free 433.7 MiB)
26/01/04 14:42:33 INFO SparkContext: Created broadcast 39 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:33 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 33 (MapPartitionsRDD[88] at
  treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:33 INFO TaskSchedulerImpl: Adding task set 33.0 with 1 tasks resource profile 0
26/01/04 14:42:33 INFO TaskSetManager: Starting task 0.0 in stage 33.0 (TID 22) (172.18.32.203,executor 0,
  partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:33 INFO TaskSetManager: Finished task 0.0 in stage 33.0 (TID 22) in 231 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:33 INFO DAGScheduler: ResultStage 33 (treeAggregate at RDDLossFunction.scala:61) finished in 310
   ms
26/01/04 14:42:33 INFO TaskSchedulerImpl: Removed TaskSet 33.0 whose tasks have all completed, from pool
26/01/04 14:42:33 INFO DAGScheduler: Job 22 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:33 INFO TaskSchedulerImpl: Canceling stage 33
26/01/04 14:42:33 INFO TaskSchedulerImpl: Killing all running tasks in stage 33: Stage finished
26/01/04 14:42:33 INFO DAGScheduler: Job 22 finished: treeAggregate at RDDLossFunction.scala:61, took 345.66557
   ms
26/01/04 14:42:33 INFO TorrentBroadcast: Destroying Broadcast(38) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:33 INFO LBFGS: Step Size: 1.000
26/01/04 14:42:33 INFO LBFGS: Val and Grad Norm: 0.449967 (rel: 0.0191) 0.0221359
26/01/04 14:42:33 INFO MemoryStore: Block broadcast_40 stored as values in memory (estimated size 104.0 B, free
   433.8 MiB)
26/01/04 14:42:33 INFO MemoryStore: Block broadcast_40_piece0 stored as bytes in memory (estimated size 186.0
  B, free 433.8 MiB)
26/01/04 14:42:33 INFO SparkContext: Created broadcast 40 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:33 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:33 INFO DAGScheduler: Got job 23 (treeAggregate at RDDLossFunction.scala:61) with 1 output
  partitions
26/01/04 14:42:33 INFO DAGScheduler: Final stage: ResultStage 34 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:33 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:33 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:33 INFO DAGScheduler: Missing parents found for ResultStage 34: List()
26/01/04 14:42:33 INFO DAGScheduler: Submitting ResultStage 34 (MapPartitionsRDD[89] at treeAggregate at
  RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:33 INFO MemoryStore: Block broadcast_41 stored as values in memory (estimated size 90.5 KiB,
  free 433.7 MiB)
26/01/04 14:42:33 INFO MemoryStore: Block broadcast_41_piece0 stored as bytes in memory (estimated size 34.9
  KiB, free 433.7 MiB)
26/01/04 14:42:33 INFO SparkContext: Created broadcast 41 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:33 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 34 (MapPartitionsRDD[89] at
  treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:33 INFO TaskSchedulerImpl: Adding task set 34.0 with 1 tasks resource profile 0
26/01/04 14:42:33 INFO TaskSetManager: Starting task 0.0 in stage 34.0 (TID 23) (172.18.32.203,executor 0,
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  partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:33 INFO TaskSetManager: Finished task 0.0 in stage 34.0 (TID 23) in 230 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:33 INFO TaskSchedulerImpl: Removed TaskSet 34.0 whose tasks have all completed, from pool
26/01/04 14:42:33 INFO DAGScheduler: ResultStage 34 (treeAggregate at RDDLossFunction.scala:61) finished in 283
   ms
26/01/04 14:42:33 INFO DAGScheduler: Job 23 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:33 INFO TaskSchedulerImpl: Canceling stage 34
26/01/04 14:42:33 INFO TaskSchedulerImpl: Killing all running tasks in stage 34: Stage finished
26/01/04 14:42:33 INFO DAGScheduler: Job 23 finished: treeAggregate at RDDLossFunction.scala:61, took
  316.234611 ms
26/01/04 14:42:33 INFO TorrentBroadcast: Destroying Broadcast(40) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:33 INFO LBFGS: Step Size: 1.000
26/01/04 14:42:33 INFO LBFGS: Val and Grad Norm: 0.445808 (rel: 0.00924) 0.00841870
26/01/04 14:42:33 INFO MemoryStore: Block broadcast_42 stored as values in memory (estimated size 104.0 B, free
   433.7 MiB)
26/01/04 14:42:33 INFO MemoryStore: Block broadcast_42_piece0 stored as bytes in memory (estimated size 186.0
  B, free 433.7 MiB)
26/01/04 14:42:33 INFO SparkContext: Created broadcast 42 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:33 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:33 INFO DAGScheduler: Got job 24 (treeAggregate at RDDLossFunction.scala:61) with 1 output
  partitions
26/01/04 14:42:33 INFO DAGScheduler: Final stage: ResultStage 35 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:33 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:33 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:33 INFO DAGScheduler: Missing parents found for ResultStage 35: List()
26/01/04 14:42:33 INFO DAGScheduler: Submitting ResultStage 35 (MapPartitionsRDD[90] at treeAggregate at
  RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:33 INFO MemoryStore: Block broadcast_43 stored as values in memory (estimated size 90.5 KiB,
  free 433.6 MiB)
26/01/04 14:42:33 INFO MemoryStore: Block broadcast_43_piece0 stored as bytes in memory (estimated size 34.9
  KiB, free 433.5 MiB)
26/01/04 14:42:33 INFO SparkContext: Created broadcast 43 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:33 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 35 (MapPartitionsRDD[90] at
  treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:33 INFO TaskSchedulerImpl: Adding task set 35.0 with 1 tasks resource profile 0
26/01/04 14:42:33 INFO TaskSetManager: Starting task 0.0 in stage 35.0 (TID 24) (172.18.32.203,executor 0,
  partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:34 INFO TaskSetManager: Finished task 0.0 in stage 35.0 (TID 24) in 176 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:34 INFO TaskSchedulerImpl: Removed TaskSet 35.0 whose tasks have all completed, from pool
26/01/04 14:42:34 INFO DAGScheduler: ResultStage 35 (treeAggregate at RDDLossFunction.scala:61) finished in 213
   ms
26/01/04 14:42:34 INFO DAGScheduler: Job 24 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:34 INFO TaskSchedulerImpl: Canceling stage 35
26/01/04 14:42:34 INFO TaskSchedulerImpl: Killing all running tasks in stage 35: Stage finished
26/01/04 14:42:34 INFO DAGScheduler: Job 24 finished: treeAggregate at RDDLossFunction.scala:61, took
  232.978429 ms
26/01/04 14:42:34 INFO TorrentBroadcast: Destroying Broadcast(42) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_44 stored as values in memory (estimated size 104.0 B, free
   433.5 MiB)
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_44_piece0 stored as bytes in memory (estimated size 186.0
  B, free 433.5 MiB)
26/01/04 14:42:34 INFO SparkContext: Created broadcast 44 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:34 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:34 INFO DAGScheduler: Got job 25 (treeAggregate at RDDLossFunction.scala:61) with 1 output
  partitions
26/01/04 14:42:34 INFO DAGScheduler: Final stage: ResultStage 36 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:34 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:34 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:34 INFO DAGScheduler: Missing parents found for ResultStage 36: List()
26/01/04 14:42:34 INFO DAGScheduler: Submitting ResultStage 36 (MapPartitionsRDD[91] at treeAggregate at
  RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_45 stored as values in memory (estimated size 90.5 KiB,
  free 433.4 MiB)
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_45_piece0 stored as bytes in memory (estimated size 34.9
  KiB, free 433.4 MiB)
26/01/04 14:42:34 INFO SparkContext: Created broadcast 45 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:34 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 36 (MapPartitionsRDD[91] at
  treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:34 INFO TaskSchedulerImpl: Adding task set 36.0 with 1 tasks resource profile 0
26/01/04 14:42:34 INFO TaskSetManager: Starting task 0.0 in stage 36.0 (TID 25) (172.18.32.203,executor 0,
  partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:34 INFO TaskSetManager: Finished task 0.0 in stage 36.0 (TID 25) in 208 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:34 INFO TaskSchedulerImpl: Removed TaskSet 36.0 whose tasks have all completed, from pool
26/01/04 14:42:34 INFO DAGScheduler: ResultStage 36 (treeAggregate at RDDLossFunction.scala:61) finished in 233
   ms
26/01/04 14:42:34 INFO DAGScheduler: Job 25 is finished. Cancelling potential speculative or zombie tasks for
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  this job
26/01/04 14:42:34 INFO TaskSchedulerImpl: Canceling stage 36
26/01/04 14:42:34 INFO TaskSchedulerImpl: Killing all running tasks in stage 36: Stage finished
26/01/04 14:42:34 INFO DAGScheduler: Job 25 finished: treeAggregate at RDDLossFunction.scala:61, took
  249.248456 ms
26/01/04 14:42:34 INFO TorrentBroadcast: Destroying Broadcast(44) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:34 INFO StrongWolfeLineSearch: Line search t: 0.5135797899124759 fval: 0.44561205703157897 rhs:
  0.44580805918287975 cdd: 5.51252968965276E-9
26/01/04 14:42:34 INFO LBFGS: Step Size: 0.5136
26/01/04 14:42:34 INFO LBFGS: Val and Grad Norm: 0.445612 (rel: 0.000440) 0.00213878
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_46 stored as values in memory (estimated size 104.0 B, free
   433.4 MiB)
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_46_piece0 stored as bytes in memory (estimated size 186.0
  B, free 433.4 MiB)
26/01/04 14:42:34 INFO SparkContext: Created broadcast 46 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:34 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:34 INFO DAGScheduler: Got job 26 (treeAggregate at RDDLossFunction.scala:61) with 1 output
  partitions
26/01/04 14:42:34 INFO DAGScheduler: Final stage: ResultStage 37 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:34 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:34 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:34 INFO DAGScheduler: Missing parents found for ResultStage 37: List()
26/01/04 14:42:34 INFO DAGScheduler: Submitting ResultStage 37 (MapPartitionsRDD[92] at treeAggregate at
  RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_47 stored as values in memory (estimated size 90.5 KiB,
  free 433.3 MiB)
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_47_piece0 stored as bytes in memory (estimated size 34.9
  KiB, free 433.3 MiB)
26/01/04 14:42:34 INFO SparkContext: Created broadcast 47 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:34 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 37 (MapPartitionsRDD[92] at
  treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:34 INFO TaskSchedulerImpl: Adding task set 37.0 with 1 tasks resource profile 0
26/01/04 14:42:34 INFO TaskSetManager: Starting task 0.0 in stage 37.0 (TID 26) (172.18.32.203,executor 0,
  partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:34 INFO TaskSetManager: Finished task 0.0 in stage 37.0 (TID 26) in 211 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:34 INFO DAGScheduler: ResultStage 37 (treeAggregate at RDDLossFunction.scala:61) finished in 235
   ms
26/01/04 14:42:34 INFO TaskSchedulerImpl: Removed TaskSet 37.0 whose tasks have all completed, from pool
26/01/04 14:42:34 INFO DAGScheduler: Job 26 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:34 INFO TaskSchedulerImpl: Canceling stage 37
26/01/04 14:42:34 INFO TaskSchedulerImpl: Killing all running tasks in stage 37: Stage finished
26/01/04 14:42:34 INFO DAGScheduler: Job 26 finished: treeAggregate at RDDLossFunction.scala:61, took
  247.937185 ms
26/01/04 14:42:34 INFO TorrentBroadcast: Destroying Broadcast(46) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:34 INFO LBFGS: Step Size: 1.000
26/01/04 14:42:34 INFO LBFGS: Val and Grad Norm: 0.445594 (rel: 4.04e-05) 0.000607130
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_48 stored as values in memory (estimated size 104.0 B, free
   433.3 MiB)
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_48_piece0 stored as bytes in memory (estimated size 186.0
  B, free 433.3 MiB)
26/01/04 14:42:34 INFO SparkContext: Created broadcast 48 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:34 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:34 INFO DAGScheduler: Got job 27 (treeAggregate at RDDLossFunction.scala:61) with 1 output
  partitions
26/01/04 14:42:34 INFO DAGScheduler: Final stage: ResultStage 38 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:34 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:34 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:34 INFO DAGScheduler: Missing parents found for ResultStage 38: List()
26/01/04 14:42:34 INFO DAGScheduler: Submitting ResultStage 38 (MapPartitionsRDD[93] at treeAggregate at
  RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_49 stored as values in memory (estimated size 90.5 KiB,
  free 433.2 MiB)
26/01/04 14:42:34 INFO MemoryStore: Block broadcast_49_piece0 stored as bytes in memory (estimated size 34.9
  KiB, free 433.2 MiB)
26/01/04 14:42:34 INFO SparkContext: Created broadcast 49 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:34 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 38 (MapPartitionsRDD[93] at
  treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:34 INFO TaskSchedulerImpl: Adding task set 38.0 with 1 tasks resource profile 0
26/01/04 14:42:34 INFO TaskSetManager: Starting task 0.0 in stage 38.0 (TID 27) (172.18.32.203,executor 0,
  partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:35 INFO TaskSetManager: Finished task 0.0 in stage 38.0 (TID 27) in 257 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:35 INFO TaskSchedulerImpl: Removed TaskSet 38.0 whose tasks have all completed, from pool
26/01/04 14:42:35 INFO DAGScheduler: ResultStage 38 (treeAggregate at RDDLossFunction.scala:61) finished in 291
   ms
26/01/04 14:42:35 INFO DAGScheduler: Job 27 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:35 INFO TaskSchedulerImpl: Canceling stage 38
26/01/04 14:42:35 INFO TaskSchedulerImpl: Killing all running tasks in stage 38: Stage finished
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26/01/04 14:42:35 INFO DAGScheduler: Job 27 finished: treeAggregate at RDDLossFunction.scala:61, took
  305.823387 ms
26/01/04 14:42:35 INFO TorrentBroadcast: Destroying Broadcast(48) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:35 INFO LBFGS: Step Size: 1.000
26/01/04 14:42:35 INFO LBFGS: Val and Grad Norm: 0.445592 (rel: 3.50e-06) 0.000100122
26/01/04 14:42:35 INFO MemoryStore: Block broadcast_50 stored as values in memory (estimated size 104.0 B, free
   433.2 MiB)
26/01/04 14:42:35 INFO MemoryStore: Block broadcast_50_piece0 stored as bytes in memory (estimated size 186.0
  B, free 433.2 MiB)
26/01/04 14:42:35 INFO SparkContext: Created broadcast 50 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:35 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:35 INFO DAGScheduler: Got job 28 (treeAggregate at RDDLossFunction.scala:61) with 1 output
  partitions
26/01/04 14:42:35 INFO DAGScheduler: Final stage: ResultStage 39 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:35 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:35 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:35 INFO DAGScheduler: Missing parents found for ResultStage 39: List()
26/01/04 14:42:35 INFO DAGScheduler: Submitting ResultStage 39 (MapPartitionsRDD[94] at treeAggregate at
  RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:35 INFO MemoryStore: Block broadcast_51 stored as values in memory (estimated size 90.5 KiB,
  free 433.1 MiB)
26/01/04 14:42:35 INFO MemoryStore: Block broadcast_51_piece0 stored as bytes in memory (estimated size 34.9
  KiB, free 433.0 MiB)
26/01/04 14:42:35 INFO SparkContext: Created broadcast 51 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:35 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 39 (MapPartitionsRDD[94] at
  treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:35 INFO TaskSchedulerImpl: Adding task set 39.0 with 1 tasks resource profile 0
26/01/04 14:42:35 INFO TaskSetManager: Starting task 0.0 in stage 39.0 (TID 28) (172.18.32.203,executor 0,
  partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:35 INFO TaskSetManager: Finished task 0.0 in stage 39.0 (TID 28) in 194 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:35 INFO TaskSchedulerImpl: Removed TaskSet 39.0 whose tasks have all completed, from pool
26/01/04 14:42:35 INFO DAGScheduler: ResultStage 39 (treeAggregate at RDDLossFunction.scala:61) finished in 215
   ms
26/01/04 14:42:35 INFO DAGScheduler: Job 28 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:35 INFO TaskSchedulerImpl: Canceling stage 39
26/01/04 14:42:35 INFO TaskSchedulerImpl: Killing all running tasks in stage 39: Stage finished
26/01/04 14:42:35 INFO DAGScheduler: Job 28 finished: treeAggregate at RDDLossFunction.scala:61, took
  229.827631 ms
26/01/04 14:42:35 INFO TorrentBroadcast: Destroying Broadcast(50) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:35 INFO LBFGS: Step Size: 1.000
26/01/04 14:42:35 INFO LBFGS: Val and Grad Norm: 0.445592 (rel: 1.35e-07) 1.53184e-05
26/01/04 14:42:35 INFO MemoryStore: Block broadcast_52 stored as values in memory (estimated size 104.0 B, free
   433.0 MiB)
26/01/04 14:42:35 INFO MemoryStore: Block broadcast_52_piece0 stored as bytes in memory (estimated size 186.0
  B, free 433.0 MiB)
26/01/04 14:42:35 INFO SparkContext: Created broadcast 52 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:35 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:35 INFO DAGScheduler: Got job 29 (treeAggregate at RDDLossFunction.scala:61) with 1 output
  partitions
26/01/04 14:42:35 INFO DAGScheduler: Final stage: ResultStage 40 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:35 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:35 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:35 INFO DAGScheduler: Missing parents found for ResultStage 40: List()
26/01/04 14:42:35 INFO DAGScheduler: Submitting ResultStage 40 (MapPartitionsRDD[95] at treeAggregate at
  RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:35 INFO MemoryStore: Block broadcast_53 stored as values in memory (estimated size 90.5 KiB,
  free 433.0 MiB)
26/01/04 14:42:35 INFO MemoryStore: Block broadcast_53_piece0 stored as bytes in memory (estimated size 34.9
  KiB, free 432.9 MiB)
26/01/04 14:42:35 INFO SparkContext: Created broadcast 53 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:35 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 40 (MapPartitionsRDD[95] at
  treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:35 INFO TaskSchedulerImpl: Adding task set 40.0 with 1 tasks resource profile 0
26/01/04 14:42:35 INFO TaskSetManager: Starting task 0.0 in stage 40.0 (TID 29) (172.18.32.203,executor 0,
  partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:35 INFO TaskSetManager: Finished task 0.0 in stage 40.0 (TID 29) in 221 ms on 172.18.32.203
  (executor 0) (1/1)
26/01/04 14:42:35 INFO TaskSchedulerImpl: Removed TaskSet 40.0 whose tasks have all completed, from pool
26/01/04 14:42:35 INFO DAGScheduler: ResultStage 40 (treeAggregate at RDDLossFunction.scala:61) finished in 253
   ms
26/01/04 14:42:35 INFO DAGScheduler: Job 29 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:35 INFO TaskSchedulerImpl: Canceling stage 40
26/01/04 14:42:35 INFO TaskSchedulerImpl: Killing all running tasks in stage 40: Stage finished
26/01/04 14:42:35 INFO DAGScheduler: Job 29 finished: treeAggregate at RDDLossFunction.scala:61, took
  271.614453 ms
26/01/04 14:42:35 INFO TorrentBroadcast: Destroying Broadcast(52) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:35 INFO LBFGS: Step Size: 1.000
26/01/04 14:42:35 INFO LBFGS: Val and Grad Norm: 0.445592 (rel: 2.73e-09) 3.63681e-06
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26/01/04 14:42:35 INFO MemoryStore: Block broadcast_54 stored as values in memory (estimated size 104.0 B, free
  432.9 MiB)
26/01/04 14:42:35 INFO MemoryStore: Block broadcast_54_piece0 stored as bytes in memory (estimated size 186.0
 B, free 432.9 MiB)
26/01/04 14:42:35 INFO SparkContext: Created broadcast 54 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:35 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:35 INFO DAGScheduler: Got job 30 (treeAggregate at RDDLossFunction.scala:61) with 1 output
 partitions
26/01/04 14:42:35 INFO DAGScheduler: Final stage: ResultStage 41 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:35 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:35 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:35 INFO DAGScheduler: Missing parents found for ResultStage 41: List()
26/01/04 14:42:35 INFO DAGScheduler: Submitting ResultStage 41 (MapPartitionsRDD[96] at treeAggregate at
 RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:35 INFO MemoryStore: Block broadcast_55 stored as values in memory (estimated size 90.5 KiB,
 free 433.0 MiB)
26/01/04 14:42:35 INFO MemoryStore: Block broadcast_55_piece0 stored as bytes in memory (estimated size 34.9
 KiB, free 432.9 MiB)
26/01/04 14:42:35 INFO SparkContext: Created broadcast 55 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:35 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 41 (MapPartitionsRDD[96] at
 treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:35 INFO TaskSchedulerImpl: Adding task set 41.0 with 1 tasks resource profile 0
26/01/04 14:42:35 INFO TaskSetManager: Starting task 0.0 in stage 41.0 (TID 30) (172.18.32.203,executor 0,
 partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:36 INFO TaskSetManager: Finished task 0.0 in stage 41.0 (TID 30) in 278 ms on 172.18.32.203
 (executor 0) (1/1)
26/01/04 14:42:36 INFO TaskSchedulerImpl: Removed TaskSet 41.0 whose tasks have all completed, from pool
26/01/04 14:42:36 INFO DAGScheduler: ResultStage 41 (treeAggregate at RDDLossFunction.scala:61) finished in 310
  ms
26/01/04 14:42:36 INFO DAGScheduler: Job 30 is finished. Cancelling potential speculative or zombie tasks for
 this job
26/01/04 14:42:36 INFO TaskSchedulerImpl: Canceling stage 41
26/01/04 14:42:36 INFO TaskSchedulerImpl: Killing all running tasks in stage 41: Stage finished
26/01/04 14:42:36 INFO DAGScheduler: Job 30 finished: treeAggregate at RDDLossFunction.scala:61, took
 334.713331 ms
26/01/04 14:42:36 INFO TorrentBroadcast: Destroying Broadcast(54) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:36 INFO LBFGS: Step Size: 1.000
26/01/04 14:42:36 INFO LBFGS: Val and Grad Norm: 0.445592 (rel: 9.99e-11) 8.32245e-07
26/01/04 14:42:36 INFO MemoryStore: Block broadcast_56 stored as values in memory (estimated size 104.0 B, free
  433.8 MiB)
26/01/04 14:42:36 INFO MemoryStore: Block broadcast_56_piece0 stored as bytes in memory (estimated size 186.0
 B, free 433.8 MiB)
26/01/04 14:42:36 INFO SparkContext: Created broadcast 56 from broadcast at RDDLossFunction.scala:57
26/01/04 14:42:36 INFO SparkContext: Starting job: treeAggregate at RDDLossFunction.scala:61
26/01/04 14:42:36 INFO DAGScheduler: Got job 31 (treeAggregate at RDDLossFunction.scala:61) with 1 output
 partitions
26/01/04 14:42:36 INFO DAGScheduler: Final stage: ResultStage 42 (treeAggregate at RDDLossFunction.scala:61)
26/01/04 14:42:36 INFO DAGScheduler: Parents of final stage: List()
26/01/04 14:42:36 INFO DAGScheduler: Missing parents: List()
26/01/04 14:42:36 INFO DAGScheduler: Missing parents found for ResultStage 42: List()
26/01/04 14:42:36 INFO DAGScheduler: Submitting ResultStage 42 (MapPartitionsRDD[97] at treeAggregate at
 RDDLossFunction.scala:61), which has no missing parents
26/01/04 14:42:36 INFO MemoryStore: Block broadcast_57 stored as values in memory (estimated size 90.5 KiB,
 free 433.7 MiB)
26/01/04 14:42:36 INFO MemoryStore: Block broadcast_57_piece0 stored as bytes in memory (estimated size 34.9
 KiB, free 433.7 MiB)
26/01/04 14:42:36 INFO SparkContext: Created broadcast 57 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:36 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 42 (MapPartitionsRDD[97] at
 treeAggregate at RDDLossFunction.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:36 INFO TaskSchedulerImpl: Adding task set 42.0 with 1 tasks resource profile 0
26/01/04 14:42:36 INFO TaskSetManager: Starting task 0.0 in stage 42.0 (TID 31) (172.18.32.203,executor 0,
 partition 0, PROCESS_LOCAL, 10250 bytes)
26/01/04 14:42:36 INFO TaskSetManager: Finished task 0.0 in stage 42.0 (TID 31) in 236 ms on 172.18.32.203
 (executor 0) (1/1)
26/01/04 14:42:36 INFO TaskSchedulerImpl: Removed TaskSet 42.0 whose tasks have all completed, from pool
26/01/04 14:42:36 INFO DAGScheduler: ResultStage 42 (treeAggregate at RDDLossFunction.scala:61) finished in 259
  ms
26/01/04 14:42:36 INFO DAGScheduler: Job 31 is finished. Cancelling potential speculative or zombie tasks for
 this job
26/01/04 14:42:36 INFO TaskSchedulerImpl: Canceling stage 42
26/01/04 14:42:36 INFO TaskSchedulerImpl: Killing all running tasks in stage 42: Stage finished
26/01/04 14:42:36 INFO DAGScheduler: Job 31 finished: treeAggregate at RDDLossFunction.scala:61, took
 270.099868 ms
26/01/04 14:42:36 INFO TorrentBroadcast: Destroying Broadcast(56) (from destroy at RDDLossFunction.scala:68)
26/01/04 14:42:36 INFO LBFGS: Step Size: 1.000
26/01/04 14:42:36 INFO LBFGS: Val and Grad Norm: 0.445592 (rel: 5.70e-12) 9.88690e-09
26/01/04 14:42:36 INFO LBFGS: Converged because gradient converged
26/01/04 14:42:36 INFO MapPartitionsRDD: Removing RDD 84 from persistence list
26/01/04 14:42:36 INFO TorrentBroadcast: Destroying Broadcast(30) (from destroy at
 LogisticRegression.scala:1009)
26/01/04 14:42:36 INFO BlockManager: Removing RDD 84
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26/01/04 14:42:36 INFO TorrentBroadcast: Destroying Broadcast(31) (from destroy at
  LogisticRegression.scala:1010)
26/01/04 14:42:37 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:42:37 INFO FileSourceStrategy: Post-Scan Filters: Set(atleastnnonnulls(6, Pclass#19, Sex#21,
  Age#22, Fare#26, Embarked#28, Survived#18))
26/01/04 14:42:37 INFO CodeGenerator: Code generated in 231.016125 ms
26/01/04 14:42:37 INFO MemoryStore: Block broadcast_58 stored as values in memory (estimated size 215.5 KiB,
  free 433.4 MiB)
26/01/04 14:42:37 INFO MemoryStore: Block broadcast_58_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.4 MiB)
26/01/04 14:42:37 INFO SparkContext: Created broadcast 58 from rdd at ClassificationSummary.scala:59
26/01/04 14:42:37 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:42:37 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:42:37 INFO FileSourceStrategy: Post-Scan Filters: Set(atleastnnonnulls(6, Pclass#19, Sex#21,
  Age#22, Fare#26, Embarked#28, Survived#18))
26/01/04 14:42:38 INFO CodeGenerator: Code generated in 265.415429 ms
26/01/04 14:42:38 INFO MemoryStore: Block broadcast_59 stored as values in memory (estimated size 215.5 KiB,
  free 433.2 MiB)
26/01/04 14:42:38 INFO MemoryStore: Block broadcast_59_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 433.2 MiB)
26/01/04 14:42:38 INFO SparkContext: Created broadcast 59 from rdd at ClassificationSummary.scala:192
26/01/04 14:42:38 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:42:39 INFO FileSourceStrategy: Pushed Filters:
26/01/04 14:42:39 INFO FileSourceStrategy: Post-Scan Filters: Set(atleastnnonnulls(6, Pclass#19, Sex#21,
  Age#22, Fare#26, Embarked#28, Survived#18))
26/01/04 14:42:39 INFO CodeGenerator: Code generated in 276.454594 ms
26/01/04 14:42:39 INFO MemoryStore: Block broadcast_60 stored as values in memory (estimated size 215.5 KiB,
  free 433.0 MiB)
26/01/04 14:42:39 INFO MemoryStore: Block broadcast_60_piece0 stored as bytes in memory (estimated size 39.1
  KiB, free 432.9 MiB)
26/01/04 14:42:39 INFO SparkContext: Created broadcast 60 from rdd at
  MulticlassClassificationEvaluator.scala:191
26/01/04 14:42:39 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost
  is considered as scanning 4194304 bytes.
26/01/04 14:42:39 INFO SparkContext: Starting job: collectAsMap at MulticlassMetrics.scala:61
26/01/04 14:42:39 INFO DAGScheduler: Registering RDD 120 (map at MulticlassMetrics.scala:52) as input to
  shuffle 9
26/01/04 14:42:39 INFO DAGScheduler: Got job 32 (collectAsMap at MulticlassMetrics.scala:61) with 1 output
  partitions
26/01/04 14:42:39 INFO DAGScheduler: Final stage: ResultStage 44 (collectAsMap at MulticlassMetrics.scala:61)
26/01/04 14:42:39 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 43)
26/01/04 14:42:39 INFO DAGScheduler: Missing parents: List(ShuffleMapStage 43)
26/01/04 14:42:39 INFO DAGScheduler: Missing parents found for ResultStage 44: List(ShuffleMapStage 43)
26/01/04 14:42:39 INFO DAGScheduler: Missing parents found for ShuffleMapStage 43: List()
26/01/04 14:42:39 INFO DAGScheduler: Submitting ShuffleMapStage 43 (MapPartitionsRDD[120] at map at
  MulticlassMetrics.scala:52), which has no missing parents
26/01/04 14:42:39 INFO MemoryStore: Block broadcast_61 stored as values in memory (estimated size 105.3 KiB,
  free 432.8 MiB)
26/01/04 14:42:39 INFO MemoryStore: Block broadcast_61_piece0 stored as bytes in memory (estimated size 39.9
  KiB, free 432.8 MiB)
26/01/04 14:42:39 INFO SparkContext: Created broadcast 61 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:39 INFO DAGScheduler: Submitting 1 missing tasks from ShuffleMapStage 43 (MapPartitionsRDD[120]
  at map at MulticlassMetrics.scala:52) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:39 INFO TaskSchedulerImpl: Adding task set 43.0 with 1 tasks resource profile 0
26/01/04 14:42:39 INFO TaskSetManager: Starting task 0.0 in stage 43.0 (TID 32) (172.18.32.200,executor 1,
  partition 0, ANY, 10239 bytes)
26/01/04 14:42:42 INFO TaskSetManager: Finished task 0.0 in stage 43.0 (TID 32) in 2375 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:42:42 INFO DAGScheduler: ShuffleMapStage 43 (map at MulticlassMetrics.scala:52) finished in 2407 ms
26/01/04 14:42:42 INFO DAGScheduler: looking for newly runnable stages
26/01/04 14:42:42 INFO DAGScheduler: running: HashSet()
26/01/04 14:42:42 INFO TaskSchedulerImpl: Removed TaskSet 43.0 whose tasks have all completed, from pool
26/01/04 14:42:42 INFO DAGScheduler: waiting: HashSet(ResultStage 44)
26/01/04 14:42:42 INFO DAGScheduler: failed: HashSet()
26/01/04 14:42:42 INFO DAGScheduler: Missing parents found for ResultStage 44: List()
26/01/04 14:42:42 INFO DAGScheduler: Submitting ResultStage 44 (ShuffledRDD[121] at reduceByKey at
  MulticlassMetrics.scala:61), which has no missing parents
26/01/04 14:42:42 INFO MemoryStore: Block broadcast_62 stored as values in memory (estimated size 4.9 KiB, free
   432.8 MiB)
26/01/04 14:42:42 INFO MemoryStore: Block broadcast_62_piece0 stored as bytes in memory (estimated size 2.8
  KiB, free 432.8 MiB)
26/01/04 14:42:42 INFO SparkContext: Created broadcast 62 from broadcast at DAGScheduler.scala:1701
26/01/04 14:42:42 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 44 (ShuffledRDD[121] at
  reduceByKey at MulticlassMetrics.scala:61) (first 15 tasks are for partitions Vector(0))
26/01/04 14:42:42 INFO TaskSchedulerImpl: Adding task set 44.0 with 1 tasks resource profile 0
26/01/04 14:42:42 INFO TaskSetManager: Starting task 0.0 in stage 44.0 (TID 33) (172.18.32.200,executor 1,
  partition 0, NODE_LOCAL, 9453 bytes)
26/01/04 14:42:42 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 9 to
  172.18.32.200:56416
```

```
26/01/04 14:42:42 INFO TaskSetManager: Finished task 0.0 in stage 44.0 (TID 33) in 123 ms on 172.18.32.200
  (executor 1) (1/1)
26/01/04 14:42:42 INFO TaskSchedulerImpl: Removed TaskSet 44.0 whose tasks have all completed, from pool
26/01/04 14:42:42 INFO DAGScheduler: ResultStage 44 (collectAsMap at MulticlassMetrics.scala:61) finished in
  152 ms
26/01/04 14:42:42 INFO DAGScheduler: Job 32 is finished. Cancelling potential speculative or zombie tasks for
  this job
26/01/04 14:42:42 INFO TaskSchedulerImpl: Canceling stage 44
26/01/04 14:42:42 INFO TaskSchedulerImpl: Killing all running tasks in stage 44: Stage finished
26/01/04 14:42:42 INFO DAGScheduler: Job 32 finished: collectAsMap at MulticlassMetrics.scala:61, took
  2594.852593 ms

Model Accuracy on Test Data: 0.7807
Accuracy saved locally to accuracy.txt
Uploading to HDFS: hdfs dfs -put -f accuracy.txt /karimi/accuracy.txt
Successfully uploaded accuracy.txt to HDFS!
26/01/04 14:42:50 INFO SparkContext: Invoking stop() from shutdown hook
26/01/04 14:42:50 INFO SparkContext: SparkContext is stopping with exitCode 0 from run at Executors.java:572.
26/01/04 14:42:50 INFO SparkUI: Stopped Spark web UI at http://raspberrypi-dml0:4040
26/01/04 14:42:50 INFO StandaloneSchedulerBackend: Shutting down all executors
26/01/04 14:42:50 INFO StandaloneSchedulerBackend$StandaloneDriverEndpoint: Asking each executor to shut down
26/01/04 14:42:50 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/01/04 14:42:50 INFO MemoryStore: MemoryStore cleared
26/01/04 14:42:50 INFO BlockManager: BlockManager stopped
26/01/04 14:42:50 INFO BlockManagerMaster: BlockManagerMaster stopped
26/01/04 14:42:50 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator
  stopped!
26/01/04 14:42:50 INFO SparkContext: Successfully stopped SparkContext (Uptime: 143159 ms)
karimi@raspberrypi-dml0:~/CA3 $ hdfs dfs -ls /karimi/
Found 2 items
-rw-r--r--   1 karimi dmls      61194 2026-01-04 14:01 /karimi/Titanic-Dataset.csv
-rw-r--r--   1 karimi dmls         16 2026-01-04 14:42 /karimi/accuracy.txt
karimi@raspberrypi-dml0:~/CA3 $ cat accuracy.txt
Accuracy: 0.7807
```

I actually got help from this chat to do the codes:
https://gemini.google.com/share/e1600f918b41

Section 3.2: Exploratory Data Analysis (EDA)

1. Survival Rate by Sex
The analysis shows a massive disparity between male and female survival rates.

- Females:
~74.2% survived.

- Males:
~18.9% survived.

```
Table for Report:
| Sex | Survival Rate (%) | Total Passengers |
| :--- | :--- | :--- |
|
Female
|
74.20%
| 314 |
|
Male
| 18.89% | 577 |
```

2. Survival Rate by Ticket Class (Pclass)
There is a clear correlation between socio-economic status and survival. First-class passengers had the highest chance of survival, while third-class passengers had the lowest.

```
Table for Report:
| Ticket Class | Survival Rate (%) | Total Passengers |
| :--- | :--- | :--- |
|
1st Class
|
62.96%
| 216 |
```

```
|
2nd Class
| 47.28% | 184 |
|
3rd Class
| 24.24% | 491 |
```

3. Average Age Analysis
The average age of survivors is slightly lower than non-survivors. This is likely due to the "women and children first" policy, where children (who have a low age) were prioritized, lowering the average age of the survivor group.

```
Table for Report:
| Group | Average Age | Count |
| :--- | :--- | :--- |
|
Survivors (1)
| 28.34 Years | 290 |
|
Non-Survivors (0)
| 30.63 Years | 424 |
```

Section 3.3 & 3.4: Classification Model Results

Model Configuration:

- Algorithm:
  Logistic Regression
- Features Used:
  Pclass, Sex (Indexed), Age, Fare, Embarked (Indexed)
- Data Split:
  80% Training (598 records), 20% Testing (114 records)

Model Performance:
The model achieved an accuracy of
78.07%
on the unseen test data. This indicates that the chosen features are strong predictors of survival.

- Accuracy:
  0.7807

HDFS File Storage Verification:
The accuracy result was successfully saved to a text file and uploaded to the HDFS storage as required.

- File Path:
  /karimi/accuracy.txt
- Content:
  Accuracy: 0.7807

Proof of Upload (Screenshot/Command Log):
(You can copy this part of your terminal output into the report as proof)

This is the compact code for this part:

```
import os
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, avg, count
from pyspark.ml.feature import StringIndexer, VectorAssembler
from pyspark.ml.classification import LogisticRegression
from pyspark.ml.evaluation import MulticlassClassificationEvaluator

```

```
# ==========================================
# 1. SETUP & CONFIGURATION
# ==========================================
APP_NAME = "Titanic_Analysis_Karimi"
HDFS_PATH = "hdfs://172.18.32.200:9000/karimi/Titanic-Dataset.csv"
HDFS_OUTPUT_DIR = "/karimi/accuracy.txt"
LOCAL_FILENAME = "accuracy.txt"

# Initialize Spark Session
spark = SparkSession.builder \
    .appName(APP_NAME) \
    .getOrCreate()

spark.sparkContext.setLogLevel("ERROR")  # Reduce log noise

# ==========================================
# 2. LOAD DATA
# ==========================================
print(f"\n[INFO] Loading data from: {HDFS_PATH} ...")
try:
    df = spark.read.csv(HDFS_PATH, header=True, inferSchema=True)
except Exception as e:
    print(f"[ERROR] Could not load file. Make sure HDFS path is correct. Error: {e}")
    spark.stop()
    exit(1)

# ==========================================
# 3. EXPLORATORY DATA ANALYSIS (EDA)
# ==========================================
print("\n" + "="*40)
print(" PART 3.2: EXPLORATORY DATA ANALYSIS")
print("="*40)

# 3.1 Survival Rate by Sex
print("\n--- Survival Rate by Sex ---")
sex_survival = df.groupBy("Sex").agg(
    (avg("Survived") * 100).alias("Survival_Rate_Percent"),
    count("PassengerId").alias("Total_Passengers")
)
sex_survival.show()

# 3.2 Survival Rate by Pclass
print("\n--- Survival Rate by Ticket Class (Pclass) ---")
class_survival = df.groupBy("Pclass").agg(
    (avg("Survived") * 100).alias("Survival_Rate_Percent"),
    count("PassengerId").alias("Total_Passengers")
).orderBy("Pclass")
class_survival.show()

# 3.3 Average Age Statistics
print("\n--- Average Age: Survivors vs Non-Survivors ---")
age_analysis = df.groupBy("Survived").agg(
    avg("Age").alias("Average_Age"),
    count("Age").alias("Count_With_Age_Data")
)
age_analysis.show()

# ==========================================
# 4. MACHINE LEARNING PIPELINE (PART 3.3)
# ==========================================
print("\n" + "="*40)
print(" PART 3.3: MACHINE LEARNING PIPELINE")
print("="*40)

# 4.1 Data Preparation
# Select features and drop missing values
required_columns = ["Pclass", "Sex", "Age", "Fare", "Embarked", "Survived"]
data = df.select(required_columns).dropna()

# 4.2 Feature Engineering (StringIndexer)
# Convert 'Sex' and 'Embarked' to numerical indices
sex_indexer = StringIndexer(inputCol="Sex", outputCol="SexIndex")
data = sex_indexer.fit(data).transform(data)

embarked_indexer = StringIndexer(inputCol="Embarked", outputCol="EmbarkedIndex")
data = embarked_indexer.fit(data).transform(data)

# 4.3 Vector Assembly
feature_cols = ["Pclass", "SexIndex", "Age", "Fare", "EmbarkedIndex"]
assembler = VectorAssembler(inputCols=feature_cols, outputCol="features")
final_data = assembler.transform(data)
```

```
# 4.4 Split Data
train_data, test_data = final_data.randomSplit([0.8, 0.2], seed=42)
print(f"Training Dataset Count: {train_data.count()}")
print(f"Test Dataset Count:     {test_data.count()}")

# 4.5 Train Model
print("\n[INFO] Training Logistic Regression Model...")
lr = LogisticRegression(featuresCol="features", labelCol="Survived")
lr_model = lr.fit(train_data)

# 4.6 Make Predictions
predictions = lr_model.transform(test_data)

# 4.7 Evaluate
evaluator = MulticlassClassificationEvaluator(
    labelCol="Survived", predictionCol="prediction", metricName="accuracy"
)
accuracy = evaluator.evaluate(predictions)
print(f"\n>>> Model Accuracy on Test Data: {accuracy:.4f}")

# ==========================================
# 5. SAVE RESULTS (PART 3.4)
# ==========================================
print("\n" + "="*40)
print(" PART 3.4: SAVING RESULTS TO HDFS")
print("="*40)

# Step A: Write to local file
with open(LOCAL_FILENAME, "w") as f:
    f.write(f"Accuracy: {accuracy:.4f}")
print(f"[INFO] Accuracy saved locally to: {LOCAL_FILENAME}")

# Step B: Upload to HDFS
# Using -f flag to overwrite if exists
upload_cmd = f"hdfs dfs -put -f {LOCAL_FILENAME} {HDFS_OUTPUT_DIR}"
print(f"[INFO] Executing: {upload_cmd}")

exit_code = os.system(upload_cmd)

if exit_code == 0:
    print("[SUCCESS] Successfully uploaded accuracy.txt to HDFS!")
else:
    print("[ERROR] Failed to upload to HDFS. Check permissions or path.")

# Stop Spark Session
spark.stop()
```

Explanation of the code:

1

Based on the assignment and our previous steps,
Question 3.1
was not about Python code. It was about using
Shell (Terminal) Commands
to interact with the Hadoop Distributed File System (HDFS).

Here is the explanation of what you did and the commands you used for Question 3.1. You should include these commands in your report.

The Concept

Before Spark (the Python code) can analyze big data, that data must live inside the distributed storage system (HDFS), not just on one local hard drive. Question 3.1 asks you to create a specific folder in that storage system and upload your file there.

The Commands Explained

1. Creating the Directory

Command:

Bash

```
hdfs dfs -mkdir -p /karimi
```

Explanation:

- hdfs
: This calls the Hadoop software.

- dfs
: Tells Hadoop we want to do a
Distributed File System
operation.

- mkdir
: Short for "Make Directory" (create a folder).

- p
: This flag is very useful. It means "no error if existing" (if the folder already exists, don't crash) and "create parents" (create any necessary sub-folders automatically).

- /karimi
: This is the folder name you chose to create inside HDFS.

2. Uploading the Data

Command:

Bash

```
hdfs dfs -put Titanic-Dataset.csv /karimi/
```

Explanation:

- put
: This is the command to
copy
a file from your
Local Linux Filesystem
(where you are logged in via SSH) into the
HDFS
.

- Titanic-Dataset.csv
: The source file on your local machine.

- /karimi/
: The destination folder inside the HDFS.

3. Verifying the Upload

Command:

Bash

```
hdfs dfs -ls /karimi/
```

Explanation:

- ls
: Short for "List". It shows the contents of a directory.

- Why we do this: To confirm the file actually arrived. In your terminal output, you saw:
rw-r--r-- 1 karimi dmls 61194 ... /karimi/Titanic-Dataset.csv
This confirms the file size (61KB) and location are correct.

Summary for your Report

For Question 3.1 in your report, you should write:

1. Objective:
To prepare the HDFS environment for analysis.

1. Step 1:
Created a directory using
hdfs dfs -mkdir -p /karimi
.

1. Step 2:
Uploaded the dataset using
hdfs dfs -put Titanic-Dataset.csv /karimi/
.

1. Step 3:
Verified the upload using
hdfs dfs -ls /karimi/
.

(This step was mandatory because the Python script in 3.2 connects to
hdfs://.../karimi/Titanic-Dataset.csv
to read this file).

2

Here is the detailed explanation of the Python code for
Part 3.2 (Exploratory Data Analysis)
.

This section of the code uses
PySpark SQL
to perform statistical analysis on the distributed data. Because the data is in HDFS, Spark distributes the calculation across the cluster nodes (though for this small dataset, it likely runs fast on one node).

1. Survival Rate by Sex

The Goal:
Find out what percentage of men vs. women survived.

Python

```
sex_survival = df.groupBy("Sex").agg(
    (avg("Survived") * 100).alias("Survival_Rate_Percent"),
    count("PassengerId").alias("Total_Passengers")
)
sex_survival.show()
```

- df.groupBy("Sex")
: This splits the data into two groups: "male" rows and "female" rows.

- .agg(...)
: Short for
Aggregate
. It allows us to perform calculations on each group.

- avg("Survived")
:
In the dataset,
Survived
is
0
(No) or
1
(Yes).
The mathematical average of a 0/1 column represents the
probability
or
rate
.
Example:
If you have [1, 1, 0, 0], the sum is 2, count is 4. Average = 0.5 (50%).

- 100
: Converts the decimal (e.g., 0.74) into a percentage (74.0).

- .alias(...)
: Renames the ugly default column name (like
(avg(Survived) * 100)
) to something readable like
"Survival_Rate_Percent"
.

- .show()
: This corresponds to an
Action
in Spark. It triggers the calculation and prints the result table to your terminal.

---

2. Survival Rate by Ticket Class (Pclass)

The Goal:
See if rich passengers (Class 1) survived more than poor passengers (Class 3).

Python

```
class_survival = df.groupBy("Pclass").agg(
    (avg("Survived") * 100).alias("Survival_Rate_Percent"),
    count("PassengerId").alias("Total_Passengers")
).orderBy("Pclass")
class_survival.show()
```

- groupBy("Pclass")
: Splits data into groups: 1, 2, and 3.

- Logic
: Same math as above (Average of 0s and 1s).

- .orderBy("Pclass")
: This sorts the final table. Without this, the rows might appear in random order (e.g., 2, 1, 3). We want them sorted 1 $\to$ 2 $\to$ 3.

---

3. Average Age: Survivors vs. Non-Survivors

The Goal:
Check if age played a role. (Did children survive more? Did older people die?)

Python

```
age_analysis = df.groupBy("Survived").agg(
    avg("Age").alias("Average_Age"),
    count("Age").alias("Count_With_Age_Data")
)
age_analysis.show()
```

- groupBy("Survived")
:
Group
0
: People who died.
Group
1
: People who survived.

- avg("Age")
: Calculates the average age for each group.

- count("Age")
: Counts how many people in that group actually had an Age listed (some rows in Titanic have null
for Age).

Summary for Report (Part 3.2)

When writing your report for 3.2, you can explain that you used
Spark SQL Aggregations
to derive these insights.

- Insight 1:
Gender was a massive factor (Females ~74% survival vs Males ~18%).

- Insight 2:
Class was a massive factor (1st Class ~62% vs 3rd Class ~24%).

- Insight 3:
Age had a smaller impact, but survivors were slightly younger on average (likely due to children being saved first).

3

Here is the detailed explanation of the code for
Part 3.3 (Machine Learning Pipeline)
.

This part represents the core "Artificial Intelligence" task. You are training a computer algorithm to look at passenger details (Age, Class, Sex, etc.) and guess whether they survived or not.

Step 1: Data Preparation

Python

```
required_columns = ["Pclass", "Sex", "Age", "Fare", "Embarked", "Survived"]
data = df.select(required_columns).dropna()
```

- Selection:
We filter the huge dataset down to only the columns that matter for prediction. Passenger Name or Ticket ID usually don't help predict survival, so we ignore them.

- dropna()
: Machine Learning algorithms generally
crash
if you feed them empty data (Nulls). This command deletes any row that has a missing value in these columns.

Step 2: Feature Engineering (StringIndexer)

Computers cannot do math with words like "male" or "female". They only understand numbers.

Python

```
sex_indexer = StringIndexer(inputCol="Sex", outputCol="SexIndex")
data = sex_indexer.fit(data).transform(data)
```

- StringIndexer
: This is a tool that translates words into numbers.
"Male" $\rightarrow$ 0
"Female" $\rightarrow$ 1

- We do this for
Sex
and
Embarked
(where the ship stopped: S, C, or Q). The result is stored in new columns called
SexIndex
and
EmbarkedIndex
.

Step 3: Vector Assembly (Crucial Spark Concept)

This is unique to Spark ML and very important to understand.

Python

```
feature_cols = ["Pclass", "SexIndex", "Age", "Fare", "EmbarkedIndex"]
assembler = VectorAssembler(inputCols=feature_cols, outputCol="features")
final_data = assembler.transform(data)
```

- Most Python libraries (like Scikit-Learn) let you feed multiple columns into a model.
Spark does not.

- Spark requires
ALL
your input numbers to be combined into a
single list

(a vector) inside
one column
.

- VectorAssembler
: It takes the 5 columns (
Pclass
,
Age
, etc.) and squashes them into one column named
"features"
.

- Example row:
Instead of
Age: 22, Sex: 0, Class: 3
, Spark sees
features: [3.0, 0.0, 22.0, ...]
.

Step 4: Splitting the Data

Python

```
train_data, test_data = final_data.randomSplit([0.8, 0.2], seed=42)
```

- We cannot test the model on the same data it studied (that would be cheating).

- 80% (Train):
Used for the model to learn patterns.

- 20% (Test):
Used to exam the model later to see how good it is.

- seed=42
: Ensures the "random" split is exactly the same every time you run the code (reproducibility).

Step 5: Training the Model

Python

```
lr = LogisticRegression(featuresCol="features", labelCol="Survived")
lr_model = lr.fit(train_data)
```

- LogisticRegression
: This is the chosen algorithm. It works well for "Yes/No" questions (Binary Classification).

- featuresCol
: We tell it where the input data is (the vector we made in Step 3).

- labelCol
: We tell it what the correct answer is (
Survived
).

- .fit(train_data)
: This is the command that starts the learning process. The computer iterates over the 598 training rows to find mathematical patterns.

Step 6: Making Predictions

Python

```
predictions = lr_model.transform(test_data)
```

- .transform(test_data)
: Now that the model is trained, we show it the
Test Data
(the 20% it has never seen).

- It looks at the features and makes a guess, storing it in a new column called
prediction
.

Summary for Report

For Part 3.3, you are building a pipeline that:

1. Cleans the data (removes Nulls).

1. Translates text to numbers (
StringIndexer
).

1. Combines features into a single vector (
VectorAssembler
).

1. Trains a Logistic Regression model on 80% of the data.

4

Here is the detailed explanation for
Part 3.4 (Saving Results)
.

This part is crucial because in a real distributed system, the computer running the code (the "Driver") is often temporary. If you don't save your results to the permanent storage (HDFS), you might lose them when the program finishes.

The Code Logic

Step 1: Write to Local Disk First

Spark (Python) cannot write a simple text file directly to HDFS easily without using specific Hadoop libraries. The standard and easiest way is to write it to the
local Linux file system
first.

Python

```
with open(LOCAL_FILENAME, "w") as f:
    f.write(f"Accuracy: {accuracy:.4f}")
```

- open(..., "w")
: Opens a file named
accuracy.txt
in
Write
mode on the computer you are currently using (

raspberrypi-dml0
).

- f.write(...)
: Writes the calculated accuracy (e.g.,
0.7807
) inside that file.

Step 2: Upload from Local to HDFS

Now that the file exists locally, we need to move it to the distributed storage. Since we are inside a Python script, we need a way to run a
Terminal Command
.

Python

```
import os
...
upload_cmd = f"hdfs dfs -put -f {LOCAL_FILENAME} {HDFS_OUTPUT_DIR}"
exit_code = os.system(upload_cmd)
```

- import os
: This library allows Python to talk to the Operating System (Linux).

- hdfs dfs -put
: The standard Hadoop command to upload files.

- f
: This stands for
Force
. It means
"If the file already exists in HDFS, overwrite it."
This is very useful if you run your code multiple times (so it doesn't crash on the second try).

- os.system(upload_cmd)
: This tells Python:
"Pause for a second, open the terminal, run this command, and then come back."

- exit_code
: If the command works, Linux returns
0
. If it fails (e.g., permission denied), it returns a non-zero number. We check this to print "Success" or "Error".

Summary for Report (Part 3.4)

In your report for this section, you should state that:

1. Method:
You calculated the accuracy using the
MulticlassClassificationEvaluator
.

1. Persistence:
You saved the result to a local file (
accuracy.txt
) and programmatically uploaded it to HDFS using
os.system
.

1. Result Location:

The final file is stored at
/karimi/accuracy.txt
in HDFS.

1. Accuracy Value:
You achieved an accuracy of
0.7807
(or 78.07%).

This completes the full explanation of the entire script! You are now fully prepared to write the report.